MTH1021 Mathematical Methods 1: Calculus in 3D
Lecture Notes 2022/2023

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Contents

Preface

This is a set of lecture notes for the last part of MTH1021 *Mathematical Methods 1*, taught over the last 8 weeks of the second semester. This part of the module can be described informally as *Calculus in 3D*. Its emphasis is on studying functions of several variables, including vector functions, and extending the ideas of differentiation and integration onto spatial derivatives and line, surface and volume integrals. We also introduce curvilinear coordinates in the plane (polar coordinates) and in three dimensions (cylindrical and spherical coordinates).

There are many books that cover these topics, often in much greater detail than this first excursion into 3D calculus requires. Typical book titles will contain the words "Vector Analysis" or "Vector Calculus". The textbooks recommended in the past for this part of Level 1 mathematics teaching at Queen's were:

1. G. Stephenson, *Mathematical Methods for Science Students* (Longman, Harlow, 1973).
2. B. Spain, *Vector Analysis* (Van Nostrand, London, 1967).
3. C. E. Weatherburn, *Advanced Vector Analysis* (Bell, London, 1924).
4. M. R. Spiegel, *Theory and Problems of Vector Analysis* (Shaum, New York, 1959).[1]
5. K. F. Riley, M. P. Hobson and S. J. Bence, *Mathematical Methods for Physics and Engineering* (Cambridge University Press, Cambridge, 2006).
6. M. L. Boas, *Mathematical Methods in the Physical Sciences*, (Wiley, Hoboken NJ, 2006)

These books are held by the Queen's Library, and you are very welcome to use them. Also, check books with similar titles and look for particular topics using the books' contents and index pages.

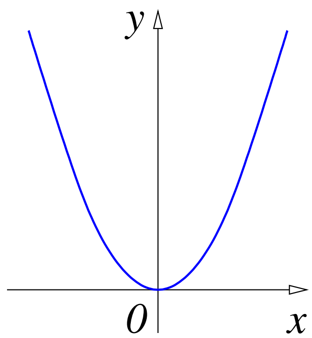
Acknowledgements
These notes were originally written by Dr Gleb Gribakin.

[1] A more recent addition is available, M. R. Spiegel, S. Lipschutz and D. Spellman, *Vector Analysis*, 2nd ed. (Schaum's Outline Series) (McGraw-Hill, New York, 2009). It can also be bought at a very reasonable price.

1 Curves in three-dimensional space

1.1 Parametric form of a curve

In this chapter we study new ways to define curves in the plane. Instead of thinking of a curve as the graph of a function, we consider a more general way of thinking of a curve as the path of a moving particle whose position is changing with time. Then each of the x- and y-coordinates of the particle's position becomes a function of a third variable t. We already know that many curves in the plane can be described using functions in Cartesian coordinates. For example, the graph of a function $y = x^2$ is a *parabola*.



A parametrisation of the function $y = x^2$ is given by

$$x = t \qquad y = t^2 \qquad -\infty < t < +\infty$$

and gives exactly the same path. The variable t is a **parameter** for the curve, specifying x and y in terms of a third variable t which often represents time. Hence we can trace out the path taken by the particle as it travels along the curves at any time t. On the other hand, a unit circle with the centre at the origin,

$$x^2 + y^2 = 1, \tag{1.1}$$

cannot be described by a single function. Solving Eq. (1.1) for y, we find

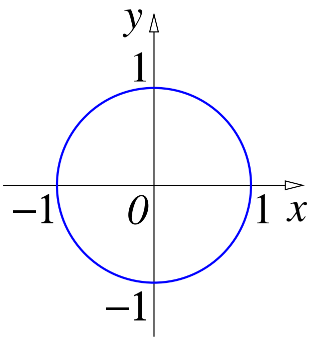
$$y = \sqrt{1 - x^2} \quad \text{or} \quad y = -\sqrt{1 - x^2}.$$

Both of these functions are defined on $-1 \leq x \leq 1$, with the first function representing the upper semicircle, and the second – the lower semicircle.

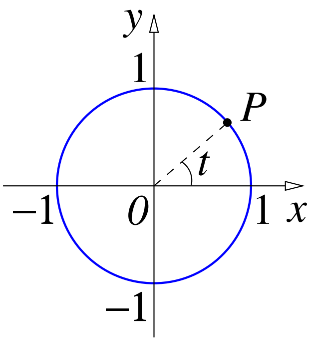
Alternatively, this circle can be described by

$$x = \cos t, \quad y = \sin t, \quad 0 \leq t \leq 2\pi, \tag{1.2}$$

i.e., by expressing the x and y coordinates as functions of some parameter t, which takes values between 0 and 2π. Indeed, x and y from Eq. (1.2) satisfy Eq. (1.1). Also, it is



easy to see that the parameter t has a simple meaning. It is the angle between the x-axis and the direction towards point $P(x, y)$ (in radians), so when t increases from 0 to 2π, the point P makes one revolution about the origin in the anticlockwise direction.[2]



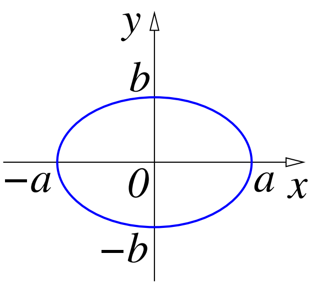
Note that to describe a circle of radius a, we simply replace (1.2) by

$$x = a\cos t, \quad y = a\sin t, \quad 0 \leq t \leq 2\pi. \tag{1.3}$$

If we now replace a by b in the expression for y,

$$x = a\cos t, \quad y = b\sin t, \quad 0 \leq t \leq 2\pi, \tag{1.4}$$

the corresponding curve will be an *ellipse* with *semiaxes* a and b.[3]



[2] If we allow t to take all real values, i.e., $t \in \mathbb{R}$, the point P will make infinitely many revolutions, i.e., each point on the circle corresponds to infinitely many values of t.

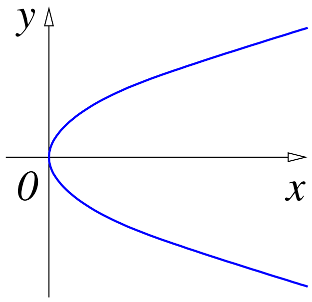
[3] For $a > b$, as shown on the diagram, a and b are known as the *semimajor* and *semiminor* axes, respectively.

In general, the equations $x = x(t)$, $y = y(t)$, where $x(t)$ and $y(t)$ are functions of a parameter t, define a curve *in parameteric form.* It can be convenient to think of t as time, and view the curve as a path followed by a point particle whose coordinates vary in time as $x(t)$ and $y(t)$.

Examples

1) The equations $x = t$, $y = t^2$, $t \in \mathbb{R}$, describe the parabola $y = x^2$, which can be seen by eliminating t.

2) The equations $x = t^2$, $y = t$, $t \in \mathbb{R}$, is the same parabola rotated clockwise by 90°. It is described by two functions: $y = \sqrt{x}$ and $y = -\sqrt{x}$, $x \geq 0$.

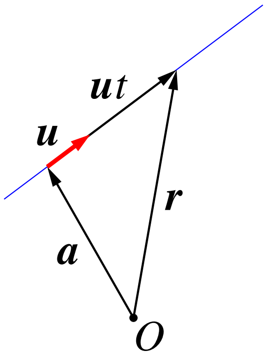


3) The equations
$$x = \frac{1-t^2}{1+t^2}, \quad y = \frac{2t}{1+t^2}, \quad -\infty < t < +\infty, \tag{1.5}$$
represent a *rational parameterisation* of the unit circle.

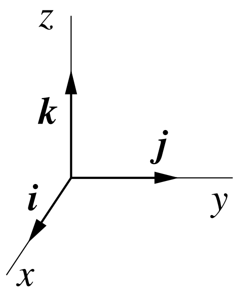
   To verify that Eqs. (1.5) describe the unit circle, first check that $x^2 + y^2 = 1$. Then examine the behaviour of x and y as functions of t: for $t \to -\infty$, $x \to -1$, $y \to 0$; for $t = -1$, $(x, y) = (0, -1)$; for $t = 0$, $(x, y) = (1, 0)$; for $t = 1$, $(x, y) = (0, 1)$; and for $t \to +\infty$, $(x, y) \to (-1, 0)$. Thus we see that as t varies from $-\infty$ to $+\infty$, the point (x, y) moves on the unit circle, starting from $(-1, 0)$ and returning to this point after one revolution in the anticlockwise direction.

4) The equations
$$x = a_x + u_x t, \quad y = a_y + u_y t, \quad t \in \mathbb{R}, \tag{1.6}$$
where a_x, a_y, u_x, u_y are constants, define a *straight line.* To see this, consider vectors $\boldsymbol{a} = (a_x, a_y)$, $\boldsymbol{u} = (u_x, u_y)$, and the *position vector* relative to the origin O, $\boldsymbol{r} = x\boldsymbol{i} + y\boldsymbol{j}$, where $\boldsymbol{i}$ and $\boldsymbol{j}$ are the unit vectors along the axes x and y, respectively. Equations (1.6) are then equivalent to the vector equation
$$\boldsymbol{r} = \boldsymbol{a} + \boldsymbol{u}t, \qquad t \in \mathbb{R}, \tag{1.7}$$
in which $\boldsymbol{a}$ is a point on the line, $\boldsymbol{u}$ gives the direction of the line, and the point $\boldsymbol{r}$ "slides" along the line as t takes all real values from $-\infty$ to $+\infty$.

Equation (1.7) can also be used to define the straight line in three dimensions. In this case we regard $\boldsymbol{r}$ as the position vector in three dimensions,
$$\boldsymbol{r} = x\boldsymbol{i} + y\boldsymbol{j} + z\boldsymbol{k}, \tag{1.8}$$



with components x, y and z, where $\boldsymbol{i}$, $\boldsymbol{j}$ and $\boldsymbol{k}$ are unit vectors in the x, y and z direction, respectively.[4] We also use the notation $\boldsymbol{r} = (x, y, z)$.



A *curve in three dimensions* can be defined by parametric equations which give the x, y and z coordinates of a point as functions of a parameter, say, t. This can be written as $x = x(t)$, $y = y(t)$, $z = z(t)$, or $\boldsymbol{r} = (x(t), y(t), z(t))$, or

$$\boldsymbol{r} = x(t)\boldsymbol{i} + y(t)\boldsymbol{j} + z(t)\boldsymbol{k}. \tag{1.9}$$

Example 1. The equations

$$x = a\cos t, \quad y = a\sin t, \quad z = bt, \quad t \in R, \tag{1.10}$$

describe a particle that moves in a circle of radius a in the xy-plane, while its z coordinate increases steadily with the rate b. This motion produces a right-handed *helix* wound about the z-axis. Here a is the radius of the helix, and b determines its pitch (i.e., period along the z-axis, which equals $2\pi b$).[5]

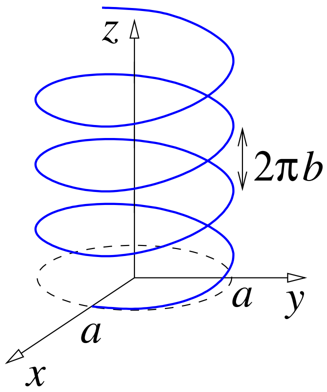
Example 2. The motion of a particle launched from the origin and affected by gravity (*projectile motion*) is described by

$$\boldsymbol{r} = \boldsymbol{v}_0 t + \frac{1}{2}\boldsymbol{g}t^2, \tag{1.11}$$

where $\boldsymbol{r}$ is the particle's position, $\boldsymbol{v}_0$ is the initial velocity, $\boldsymbol{g}$ is the acceleration due to gravity (vertically downwards), and t is time.

[4] Note the direction of the positive z-axis relative to the xy-plane: when viewed from the "top" of the z-axis, rotation from the x- to the y-axis is anticlockwise.

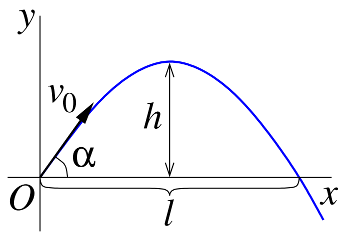
[5] Probably the most famous example of a helix is given by DNA molecules which form a double helix, each with $a = 10$ Å and $2\pi b = 34$ Å (Å is angstrom, 1 Å $= 10^{-10}$ m).



Let us choose the xy-plane so that it contains vectors $\boldsymbol{v}_0$ and $\boldsymbol{g}$, and the motion is confined to this plane. If the y-axis is vertically up (in the direction opposite to $\boldsymbol{g}$), and the initial velocity makes angle α with the horizontal x-axis, we have

$$\boldsymbol{v}_0 = v_0 \cos\alpha \boldsymbol{i} + v_0 \sin\alpha \boldsymbol{j}, \quad \boldsymbol{g} = -g\boldsymbol{j}.$$

The vector equation (1.11) can now be written in components as



$$x = v_0 \cos\alpha\, t, \tag{1.12}$$

$$y = v_0 \sin\alpha\, t - \frac{1}{2} g t^2. \tag{1.13}$$

The curve described by either Eq. (1.11) or Eqs. (1.12) and (1.13) is a parabola. To see this, we use Eq. (1.12) to find $t = x/(v_0 \cos\alpha)$, and eliminating t from (1.13), obtain

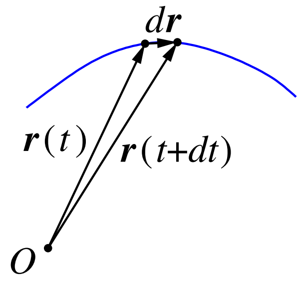
$$y = x \tan\alpha - \frac{g x^2}{2 v_0^2 \cos^2\alpha}, \tag{1.14}$$

which is a parabola with downward branches. From Eq. (1.14), it is straightforward to find the maximum height h and the horizontal range l.

1.2 Element of length and arc length of a curve

Consider two points on the curve, $\boldsymbol{r}(t)$ and $\boldsymbol{r}(t+dt)$, where dt is the change in the value of t. For small dt, the two points are close to each other. They are separated by a small displacement

$$d\boldsymbol{r} = \boldsymbol{r}(t+dt) - \boldsymbol{r}(t),$$



along the curve. The *element of length* of the curve is the length of $d\boldsymbol{r}$,

$$ds = |d\boldsymbol{r}| = \left|\frac{d\boldsymbol{r}}{dt}\right| dt = |\boldsymbol{r}'(t)|dt, \tag{1.15}$$

where $\boldsymbol{r}'(t)$ is the derivative of $\boldsymbol{r}(t)$ with respect to t.

For $\boldsymbol{r}$ given in Cartesian components, Eq. (1.8), $d\boldsymbol{r} = dx\boldsymbol{i} + dy\boldsymbol{j} + dz\boldsymbol{k}$, so that

$$\frac{d\boldsymbol{r}}{dt} = \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt}\right), \tag{1.16}$$

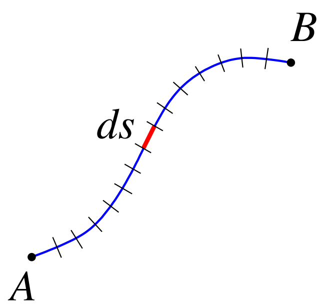
and

$$\left|\frac{d\boldsymbol{r}}{dt}\right| = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}. \tag{1.17}$$

Accordingly, the element of length is

$$ds = |d\boldsymbol{r}| = \sqrt{dx^2 + dy^2 + dz^2} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt\,. \tag{1.18}$$

If we now consider a segment of a curve between points A $(t = a)$ and B $(t = b)$, its



length can be found by dividing the curve into small length elements ds, and summing them. This is equivalent to finding the integral

$$s_{AB} = \int_A^B ds = \int_a^b |\boldsymbol{r}'(t)|dt = \int_a^b \sqrt{[x'(t)]^2 + [y'(t)]^2 + [z'(t)]^2}\, dt\,, \tag{1.19}$$

which gives the *arc length* of the curve.

Example: Find the arc length of one turn of the helix (1.10).

From Eqs. (1.10),

$$x'(t) = -a\sin t, \quad y'(t) = a\cos t, \quad z'(t) = b,$$

which gives the length element

$$ds = \sqrt{(-a\sin t)^2 + (a\cos t)^2 + b^2}\, dt = \sqrt{a^2+b^2}\, dt.$$

Integrating this expression over one turn of the helix, e.g., between $t = 0$ and 2π, we obtain

$$s = \int_0^{2\pi} \sqrt{a^2+b^2}\, dt = 2\pi\sqrt{a^2+b^2}.$$

This results reminds one of Pythagoras's theorem. Indeed, if the helix were drawn on a cylinder of radius a, which was then cut lengthwise and unrolled, the length of one turn would be given by the hypotenuse of a right-angle trangle with short sides $2\pi a$ and $2\pi b$.

1.3 Tangent, normal and binormal vectors

The behaviour of a three-dimensional curve near a point is characterised by three mutually orthogonal unit vectors, called tangent, normal and binormal (the Frenet-Serret frame), and two scalars known as curvature and torsion.

Tangent vector $\boldsymbol{T}$.
For a curve described by $\boldsymbol{r} = \boldsymbol{r}(t)$, the *tangent* vector $\boldsymbol{T}$ is a unit vector in the direction of $d\boldsymbol{r}$ (see Sec. 1.2). It is given by[6]

$$\boldsymbol{T} = \frac{d\boldsymbol{r}}{ds} = \frac{1}{|\boldsymbol{r}'(t)|}\frac{d\boldsymbol{r}}{dt} = \frac{\boldsymbol{r}'(t)}{|\boldsymbol{r}'(t)|}, \tag{1.20}$$

where we used Eq. (1.15). As follows from the definition, $\boldsymbol{T}$ is tangential to the curve. Before proceeding, let us prove a useful lemma.

Lemma. Consider a vector $\boldsymbol{a}$ which depends on some parameter, say, t. If $\boldsymbol{a}$ has a constant length, then the vector $d\boldsymbol{a}/dt$ is perpendicular to $\boldsymbol{a}$.

Proof. The squared length of vector $\boldsymbol{a}$ can be written as a scalar product[7] of $\boldsymbol{a}$ with itself, i.e., $\boldsymbol{a}\cdot\boldsymbol{a} = a^2 = \text{const}$. Differentiating this with respect to t, we find

$$\frac{d}{dt}(\boldsymbol{a}\cdot\boldsymbol{a}) = \frac{d\boldsymbol{a}}{dt}\cdot\boldsymbol{a} + \boldsymbol{a}\cdot\frac{d\boldsymbol{a}}{dt} = 2\,\frac{d\boldsymbol{a}}{dt}\cdot\boldsymbol{a} = 0,$$

which shows that $d\boldsymbol{a}/dt$ and $\boldsymbol{a}$ are perpendicular.

Normal vector $\boldsymbol{N}$ and curvature κ.
The unit *normal* vector $\boldsymbol{N}$ is defined by

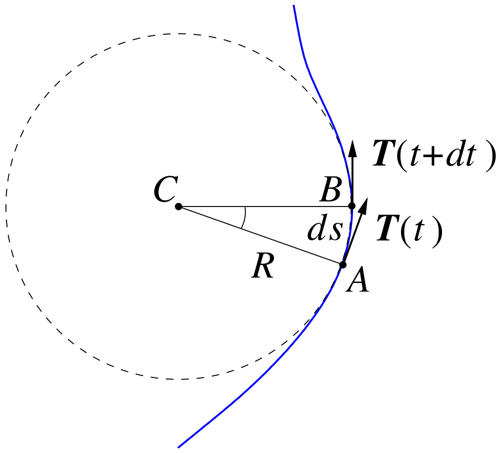
$$\frac{d\boldsymbol{T}}{ds} = \kappa\boldsymbol{N}, \tag{1.21}$$

[6] Recall that a unit vector in the direction of a given vector, say, $\boldsymbol{a}$, is found by dividing it by its length $|\boldsymbol{a}| \equiv a$, i.e., as $\boldsymbol{a}/a$.
[7] The scalar product of vectors $\boldsymbol{a}$ and $\boldsymbol{b}$ is $\boldsymbol{a}\cdot\boldsymbol{b} = ab\cos\theta$ (θ is the angle between $\boldsymbol{a}$ and $\boldsymbol{b}$).

where κ is the *curvature*. Since $|\boldsymbol{T}| = 1$, the above lemma tells us that $d\boldsymbol{T}/ds$ is perpendicular to $\boldsymbol{T}$. This means that $\boldsymbol{N}$ from Eq. (1.21) is also perpendicular to $\boldsymbol{T}$. Further, since $\boldsymbol{N}$ is a unit vector, the curvature is given by

$$\kappa = |d\boldsymbol{T}/ds|. \tag{1.22}$$

To understand the meaning of the curvature, we introduce the *osculating circle* that



approximates the arc of a curve near a given point $\boldsymbol{r}(t)$ in the closest possible way (from Latin *osculari* – 'to kiss'). We consider two nearby points on the curve: A that corresponds to $\boldsymbol{r}(t)$, and B that corresponds to $\boldsymbol{r}(t + dt)$, and show the tangent vectors $\boldsymbol{T}(t)$ and $\boldsymbol{T}(t+dt)$ at these points. We also show two line segments, CA and CB, whose lengths are equal to the radius R of the osculating circle, C being its centre. Since $\boldsymbol{T}(t)$ and $\boldsymbol{T}(t + dt)$ are perpendicular to the radii CA and CB, respectively, the isosceles triangle formed by the vectors $\boldsymbol{T}(t)$, $\boldsymbol{T}(t + dt)$ and $d\boldsymbol{T} = \boldsymbol{T}(t + dt) - \boldsymbol{T}(t)$ is similar to the triangle ABC.

Hence, $|d\boldsymbol{T}|/|\boldsymbol{T}| = ds/R$, as ratios of corresponding sides, which shows that

$$\frac{|d\boldsymbol{T}|}{ds} = \frac{1}{R}. \tag{1.23}$$

Comparing this with Eq. (1.22), we see that

$$\kappa = 1/R, \tag{1.24}$$

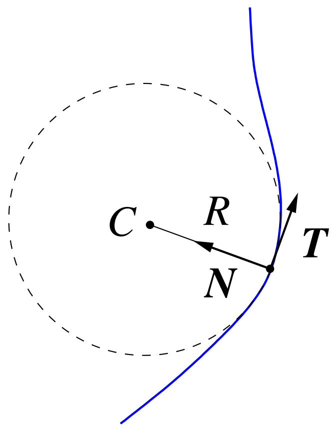
i.e., the curvature κ is the reciprocal of the radius of the osculating circle R, known as the *radius of curvature*. The direction of $\boldsymbol{N}$, perpendicular to $\boldsymbol{T}$, is towards the centre of the osculating circle, known as the centre of curvature.

Note that using Eq. (1.15), we can write the derivative of $\boldsymbol{T}$ in Eq. (1.21) as

$$\frac{d\boldsymbol{T}}{ds} = \frac{1}{|\boldsymbol{r}'(t)|}\frac{d\boldsymbol{T}}{dt}, \tag{1.25}$$

so the unit vector $\boldsymbol{N}$ can be found as

$$\boldsymbol{N} = \frac{d\boldsymbol{T}/dt}{|d\boldsymbol{T}/dt|}. \tag{1.26}$$



Binormal vector $\boldsymbol{B}$ and torsion τ.
The third unit vector, the *binormal* $\boldsymbol{B}$, is constructed to be perpendicular to both $\boldsymbol{T}$ and $\boldsymbol{N}$, using the cross product

$$\boldsymbol{B} = \boldsymbol{T} \times \boldsymbol{N}. \tag{1.27}$$

For its derivative with respect to the length element, we have

$$\frac{d\boldsymbol{B}}{ds} = \frac{d}{ds}(\boldsymbol{T} \times \boldsymbol{N}) = \underbrace{\frac{d\boldsymbol{T}}{ds} \times \boldsymbol{N}}_{=0} + \boldsymbol{T} \times \frac{d\boldsymbol{N}}{ds} = \boldsymbol{T} \times \frac{d\boldsymbol{N}}{ds}, \tag{1.28}$$

where the first term vanished because $d\boldsymbol{T}/ds$ is parallel to $\boldsymbol{N}$, see Eq. (1.21). Since $d\boldsymbol{N}/ds$ is perpendicular to $\boldsymbol{N}$ (by the lemma), it lies in the plane of vectors $\boldsymbol{T}$ and $\boldsymbol{B}$, and the cross product on the right-hand side of (1.28) is perpendicular to this plane, e.g., along $\boldsymbol{N}$. Hence, we can write

$$\frac{d\boldsymbol{B}}{ds} = -\tau \boldsymbol{N}, \tag{1.29}$$

where τ is the *torsion*, and the minus sign is introduced by convention[8].

For a plane curve, i.e., a curve that lies in a plane, both $\boldsymbol{T}$ and $\boldsymbol{N}$ lie in this plane, and $\boldsymbol{B}$ is perpendicular to it. In this case, $\boldsymbol{B} = \text{const}$ and $\tau = 0$.

When the torsion is nonzero, the curve is not a plane curve. The magnitude of the torsion shows how quickly a curve "twists out" of a plane.

The helix (1.10) from Example 1 in Sec. 1.1 is not a plane curve, while the parabola (1.11) in Example 2 on the projectile motion, is a plane curve.

By taking the derivative of $\boldsymbol{N} = \boldsymbol{B} \times \boldsymbol{T}$, one can also show that

$$\frac{d\boldsymbol{N}}{ds} = -\kappa \boldsymbol{T} + \tau \boldsymbol{B}. \tag{1.30}$$

[8] This convention makes the torsion of a right-handed helix positive.

2 Functions of several variables

2.1 Basic ideas

The notation $y = f(x)$ is used to indicate that the variable y depends on the single real variable x and the domain of such a function is a set of real numbers. Many quantities, however, can be regarded as depending on more than one real variable and are thus defined to be functions of more than one variable. For example, the volume of a circular cylinder of radius r and height h is given by $V = \pi r^2 h$ and we say that V is a function of the two variables r and h. Hence we could write $V = f(r, h)$ where f is a function of two variables having as domain the set of points in the $rh-$plane with coordinates (r, h) satisfying $r \geq 0$ and $h \geq 0$.

Definition. A function of two variables (say, x and y) is a rule which supplies a single value (z) for each pair (x, y) from a certain set (*domain*).

This is written as $z = f(x, y)$, where x and y are the arguments or independent variables.

The "rule" is often given by means of a formula, e.g.,

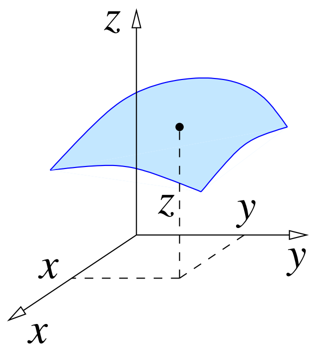
$$f(x, y) = \sqrt{x^2 + y^2}. \tag{2.1}$$

This function gives the distance from the origin to the point (x, y).

Functions can also be defined by other means, e.g.,

$$f(x, y) = \begin{cases} 1, & x > y \\ 0, & x = y \\ -1, & x < y \end{cases}.$$

A function of two variables $z = f(x, y)$ represents a surface. In Cartesian coordinates, z gives the height of the point above the xy-plane.



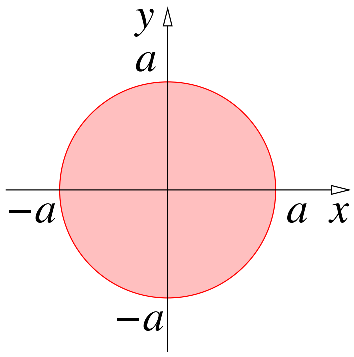
<u>Example.</u> The function

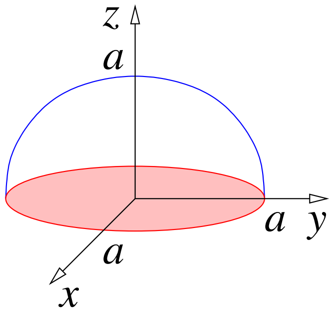
$$f(x, y) = \sqrt{a^2 - x^2 - y^2} \tag{2.2}$$

is defined for $x^2 + y^2 \leq a^2$, i.e.,

$$\sqrt{x^2 + y^2} \leq a,$$

so the domain of this function is a disc[9] of radius a centred on the origin.





The shape of the surface described by (2.2) is a "dome", i.e., a hemisphere with a maximum $z = a$ at the point $(0, 0)$.

To describe the full sphere one will need two functions, $z = \sqrt{a^2 - x^2 - y^2}$ and $z = -\sqrt{a^2 - x^2 - y^2}$. Alternatively, this can be done by setting

$$\begin{aligned} x &= a \sin\theta \cos\phi, \\ y &= a \sin\theta \sin\phi, \\ z &= a \cos\theta, \end{aligned} \tag{2.3}$$

with $0 \leq \theta \leq \pi$ and $0 \leq \phi \leq 2\pi$ being two parameters.

Exercise: verify that equations (2.3) satisfy $x^2 + y^2 + z^2 = a^2$.

Similarly, one can consider functions of more than two variables.

Example. The temperate in this room is described by $T = f(x, y, z)$.

In fact, the temperature also changes with time t, so it is a function of four variables, $T = f(x, y, z, t)$.

A function of n variables can be written as

$$f(x_1, x_2, \ldots, x_n).$$

[9] A disc is the region in a plane bounded by a circle.

Examples of domains.

1. For $z = \sqrt{x+y}$ the domain is $x + y \geq 0$, or $y \geq -x$, i.e., all points above the line $y = -x$, including the line itself.
2. For $z = \ln(x - y)$ the domain is $x - y > 0$, or $y < x$, i.e., all points below the line $y = x$, but not including the line.
3. For $u = \sqrt{x} + \sqrt{y} + \sqrt{z}$ the domain is $x \geq 0$, $y \geq 0$, $z \geq 0$, i.e., the first octant, including the faces that bound it.

You can play about with plotting these functions of several variables using the GeoGebra calculator suite at **geogebra.org**. Visualising these plots can be useful for effective learning.

2.2 Limits and continuity

This section deals with limits and continuity of functions of more than one variable. The definitions are the same as for functions of a single variable except that there are now more variables to consider. Lets recap on the definition for a function of a single variable

Definition (single varaiable). A number L is said to be the limit of $f(x)$ at the point $x = a$, if for any $\varepsilon > 0$, there exists a number δ such that

$$0 < \sqrt{(x-a)^2} < \delta \quad \Rightarrow \quad |f(x) - L| < \varepsilon.$$

In this case we write

$$\lim_{x \to a} f(x) = L.$$

This formal definition of a limit does not tell you how to find the limit of the function, but it does enable you to verify that a suspected limit is correct. If the limit exists it is unique in that the function approaches the same finite number whether it approaches from the left or the right. This definition can be extended to a function of several variables.

Definition (several varaiables). A number L is said to be the limit of $f(x, y)$ at the point $x = a$, $y = b$, if for any $\varepsilon > 0$, there is δ such that

$$0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta \quad \Rightarrow \quad |f(x, y) - L| < \varepsilon.$$

In this case we write

$$\lim_{\substack{x \to a \\ y \to b}} f(x, y) = L.$$

Again the limit is unique and the function $f(x, y)$ approaches the same finite number L regardless of the path it takes to get there. In other words (x, y) can approach (a, b) along any curve that lies in the domain of f. Note that this limit is not the same as

$$\lim_{x \to a} \lim_{y \to b} f(x, y) \quad \text{or} \quad \lim_{y \to b} \lim_{x \to a} f(x, y).$$

Example. Consider the function $f(x,y) = \dfrac{x^2 - y^2}{x^2 + y^2}$ defined everywhere except $(0,0)$. We have

$$\lim_{x\to 0}\lim_{y\to 0}\frac{x^2 - y^2}{x^2 + y^2} = \lim_{x\to 0} 1 = 1,$$

while

$$\lim_{y\to 0}\lim_{x\to 0}\frac{x^2 - y^2}{x^2 + y^2} = \lim_{y\to 0}(-1) = -1.$$

Hence, this function does not have a limit for $(x,y) \to (0,0)$.

What does it do then near the origin then? If you cannot make any headway with limits in cartesian coordinates, try changing to polar coordinates. Let us investigate the behavior of this function using polar coordinates r and θ, in which $x = r\cos\theta$ and $y = r\sin\theta$. We have

$$f(x,y) = \frac{r^2(\cos^2\theta - \sin^2\theta)}{r^2(\cos^2\theta + \sin^2\theta)} = \cos 2\theta.$$

As we go around the origin in a circle, starting on the x-axis, this function decreases from unity, becomes zero (for $\theta = \pi/4$), then -1 (for $\theta = \pi/2$), etc. It is equal to $+1$ everywhere on the x-axis (except at the origin), and to -1 on the y-axis, so the "limit" one finds depends on the direction from which one approaches the point $(0,0)$. Hence, this function does not have the limit in the sense of the above definition.

Definition. A function $f(x,y)$ is continuous at the point (a,b) if

$$\lim_{\substack{x\to a\\ y\to b}} f(x,y) = f(a,b).$$

2.3 Partial derivatives

Definition. For a function $f(x,y)$, the limit

$$\frac{\partial f}{\partial x} = \lim_{\Delta x\to 0}\frac{f(x+\Delta x, y) - f(x,y)}{\Delta x} \equiv f'_x(x,y), \tag{2.4}$$

is the partial derivative of $f(x,y)$ with respect to x. Similarly,

$$\frac{\partial f}{\partial y} = \lim_{\Delta y\to 0}\frac{f(x, y+\Delta y) - f(x,y)}{\Delta y} \equiv f'_y(x,y), \tag{2.5}$$

is the partial derivative with respect to y.

When taking a partial derivative with respect to a given variable, the other variables are regarded as constants.

Example. For $z = \ln\tan\dfrac{x}{y}$,

$$\frac{\partial z}{\partial x} = \frac{2}{y\sin\dfrac{2x}{y}},$$

$$\frac{\partial z}{\partial y} = -\frac{2x}{y^2 \sin \dfrac{2x}{y}},$$

where we used the chain rule and the double-angle formula $\sin 2\theta = 2 \sin\theta \cos\theta$.

The same idea can be applied to functions of more than two variables.

Example. For $u = x^3y^2z + 2x - 3y + z + 5$,

$$\frac{\partial u}{\partial x} = 3x^2y^2z + 2,$$

$$\frac{\partial u}{\partial y} = 2x^3yz - 3,$$

$$\frac{\partial u}{\partial z} = x^3y^2 + 1.$$

2.4 Higher-order partial derivatives

One can take partial derivatives of partial derivatives to obtain higher-order partial derivatives.

For example, for a function $f(x, y)$, we find four second-order partial derivatives

$$\frac{\partial}{\partial x}\frac{\partial f}{\partial x} = \frac{\partial^2 f}{\partial x^2} \equiv f''_{xx}(x, y), \tag{2.6}$$

$$\frac{\partial}{\partial y}\frac{\partial f}{\partial x} = \frac{\partial^2 f}{\partial y \partial x} \equiv f''_{xy}(x, y), \tag{2.7}$$

$$\frac{\partial}{\partial x}\frac{\partial f}{\partial y} = \frac{\partial^2 f}{\partial x \partial y} \equiv f''_{yx}(x, y), \tag{2.8}$$

$$\frac{\partial}{\partial y}\frac{\partial f}{\partial y} = \frac{\partial^2 f}{\partial y^2} \equiv f''_{yy}(x, y). \tag{2.9}$$

The derivatives (2.7) and (2.8) are *mixed* partial derivatives.

Example. For $f(x, y) = \ln(x^2 + y^2)$,

$$\frac{\partial f}{\partial x} = \frac{2x}{x^2 + y^2}, \qquad \frac{\partial f}{\partial y} = \frac{2y}{x^2 + y^2},$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{2(y^2 - x^2)}{(x^2 + y^2)^2}, \qquad \frac{\partial^2 f}{\partial y^2} = \frac{2(x^2 - y^2)}{(x^2 + y^2)^2}, \tag{2.10}$$

$$\frac{\partial^2 f}{\partial y \partial x} = -\frac{4xy}{(x^2 + y^2)^2}, \qquad \frac{\partial^2 f}{\partial x \partial y} = -\frac{4xy}{(x^2 + y^2)^2}, \tag{2.11}$$

Note that the mixed partial derivatives above are equal.

It can be shown in general that if the mixed partial derivatives of any order are continuous, then they are independent of the order of differentiation, e.g.

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}.$$

Similarly, one can find higher-order partial derivatives for a function with more variables, e.g., for $u = f(x, y, x)$:

$$\frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial x \partial y}, \ldots, \frac{\partial^2 f}{\partial z^2}, \quad \frac{\partial^3 f}{\partial x^3}, \frac{\partial^3 f}{\partial x^2 \partial y}, \frac{\partial^3 f}{\partial x \partial y^2}, \ldots, \frac{\partial^3 f}{\partial x \partial y \partial z}, \ldots, \frac{\partial^3 f}{\partial z^3}.$$

Exercise. How many different second-order partial derivatives are there for a function of three variables? Different third-order derivatives?

2.5 Total differential of a function

Recall: for a function of one variable, $f(x)$, the differential is $df = f'(x)dx$.

For a function of two variables $z = f(x, y)$, we want to do something similar to this. We have said that the equation represents a surface and that the derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ at a point are the slopes of the two tangent lines to the surface in the x and y directions at that point. The symbol $\Delta x = dx$ and $\Delta y = dy$ represent changes in the independent variables x and y. The quantity Δf or indeed Δz means the corresponding change in f along the surface. Consider this change in the function of two variables,

$$\Delta f = f(x + \Delta x, y + \Delta y) - f(x, y).$$

Definition. The *total differential* is the main, linear part of Δf,

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy. \tag{2.12}$$

For a function of n variables $f(x_1, x_2, \ldots, x_n)$,

$$df = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i} dx_i. \tag{2.13}$$

In three dimensions, for $u = f(x, y, z)$,

$$du \equiv df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz. \tag{2.14}$$

Derivative of a composite function.

Consider a function $f(x, y, z)$ in which x, y and z depend on a parameter t, i.e., $x = x(t)$, $y = y(t)$ and $z = z(t)$, so that f depends on t. Using Eq. (2.14), we find

$$\frac{df}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt} + \frac{\partial f}{\partial z}\frac{dz}{dt}. \tag{2.15}$$

This can be viewed as the chain rule for a function of several variables.

2.6 Gradient vector and directional derivative

Equation (2.14) has the form of a scalar product of two vectors. The vector with components $\partial f/\partial x$, $\partial f/\partial y$ and $\partial f/\partial z$ is denoted by

$$\nabla f = \frac{\partial f}{\partial x}\boldsymbol{i} + \frac{\partial f}{\partial y}\boldsymbol{j} + \frac{\partial f}{\partial z}\boldsymbol{k}, \tag{2.16}$$

and is called the *gradient* of the function. The other vector is

$$d\boldsymbol{r} = dx\boldsymbol{i} + dy\boldsymbol{j} + dz\boldsymbol{k}.$$

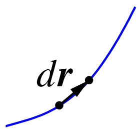
The total differential of f can now be written as $df = \nabla f \cdot d\boldsymbol{r}$.

Here

$$\nabla = \boldsymbol{i}\frac{\partial}{\partial x} + \boldsymbol{j}\frac{\partial}{\partial y} + \boldsymbol{k}\frac{\partial}{\partial z} \tag{2.17}$$

is the *differential operator* called *nabla*[10].

<u>Directional derivative.</u>



Let us consider the change in the function f associated with a small displacement $d\boldsymbol{r}$ along a curve, $df = \nabla f \cdot d\boldsymbol{r}$. The associated rate of change is

$$\frac{df}{ds} = \nabla f \cdot \frac{d\boldsymbol{r}}{ds} \tag{2.18}$$

where $ds = |d\boldsymbol{r}|$ and $d\boldsymbol{r}/ds \equiv \boldsymbol{T}$ is the unit tangent vector (see Sec. 1.3).

The rate of change of a function along a curve or line is known as the *directional derivative*, and is denoted $\partial f/\partial s$[11]. According to Eq. (2.18), it is given by

$$\frac{\partial f}{\partial s} = \nabla f \cdot \boldsymbol{T} = |\nabla f| \cos\theta, \tag{2.19}$$

where $\boldsymbol{T}$ is the unit vector in the direction of the line, and θ is the angle between ∇f and $\boldsymbol{T}$.

Equation (2.19) shows that the directional derivative takes its maximum value $|\nabla f|$ when $\boldsymbol{T}$ is in the direction of the gradient vector ∇f (i.e., $\theta = 0$). This is the direction of the fastest increase of the function f.

[10] This operator was introduced in 1837 by the great Irish mathematician and physicist William Rowan Hamilton (1805-1865). The symbol ∇ was adopted by the Scottish mathematical physicist Peter Guthrie Tait around 1870, "nabla" being the Hellenistic Greek word νάβλα for a Phoenician harp. A similar Hebrew word "nevel" also means "harp".

[11] Here the partial derivative notation is used, to make this similar to $\partial f/\partial x$, $\partial f/\partial y$, etc.

2.7 Taylor's expansion

The Taylor expansion for a function of $f(x)$ about the point $x = a$ is

$$f(x) = f(a) + f'(a)(x-a) + \frac{1}{2}f''(a)(x-a)^2 + \ldots + \frac{1}{n!}f^{(n)}(a)(x-a)^n + \ldots, \quad (2.20)$$

where $f^{(n)}(a)$ is the nth derivative of $f(x)$ evaluated at point a.

Letting $x = a + h$, we can write it as an expansion in powers of h,

$$f(a+h) = f(a) + f'(a)h + \frac{1}{2}f''(a)h^2 + \ldots + \frac{1}{n!}f^{(n)}(a)h^n + \ldots. \quad (2.21)$$

To write this expansion in a compact form, let us introduce the *operator*

$$D = h\frac{d}{dx}, \quad (2.22)$$

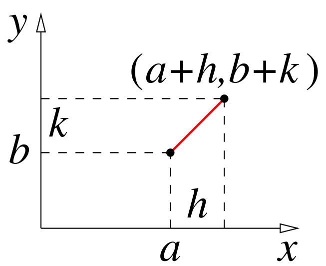
that differentiates the function to its right, and multiplies the result by h. Equation (2.21) can now be written as

$$f(a+h) = f(a) + Df + \frac{1}{2}D^2 f + \frac{1}{3!}D^3 f + \ldots + \frac{1}{n!}D^n f + \ldots = \sum_{n=0}^{\infty}\frac{1}{n!}D^n f, \quad (2.23)$$

where its is assumed that all the derivatives are evaluated at $x = a$. The equivalence (2.23) and (2.21) can easily be verified by seen that

$$Df = h\frac{df}{dx} = hf'(a), \quad D^2 f = h\frac{d}{dx}h\frac{df}{dx} = h^2 f''(a), \quad \ldots,$$

h being a constant here.



To expand a function of two variables $f(x, y)$ about the point (a, b), we replace D from Eq. (2.22) by

$$D = h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}, \quad (2.24)$$

where h and k are the steps along the x and y axes that take one from point (a, b) to $(a+h, b+k)$[12].

The Taylor expansion retains its form (2.23), and we have

$$f(a+h, b+k) = f(a, b) + Df + \frac{1}{2}D^2 f + \frac{1}{3!}D^3 f + \ldots + \frac{1}{n!}D^n f + \ldots, \tag{2.25}$$

where all the partial derivatives are evaluated at (a, b).

Expanding the terms, we find

$$\begin{aligned}
Df &= h\frac{\partial f}{\partial x} + k\frac{\partial f}{\partial y}, \\
D^2 f &= \left(h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right)^2 f = h^2\frac{\partial^2 f}{\partial x^2} + 2hk\frac{\partial^2 f}{\partial x \partial y} + k^2\frac{\partial^2 f}{\partial y^2}, \\
D^3 f &= \left(h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right)^3 f = h^3\frac{\partial^3 f}{\partial x^3} + 3h^2 k\frac{\partial^3 f}{\partial^2 x \partial y} + 3hk^2\frac{\partial^3 f}{\partial x \partial y^2} + k^3\frac{\partial^3 f}{\partial y^3},
\end{aligned}$$

or in general,[13]

$$D^n f = \left(h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right)^n f = \sum_{m=0}^{n} \binom{n}{m} h^{n-m} k^m \frac{\partial^n f}{\partial x^{n-m} \partial y^m}.$$

Explicitly, the Taylor expansion (2.25) up to second order is

$$f(a+h, b+k) = f(a, b) + f'_x h + f'_y k + \frac{1}{2}\left[f''_{xx} h^2 + 2f''_{xy} hk + f''_{yy} k^2\right] + \ldots, \tag{2.26}$$

where all the derivatives are evaluated at (a, b).

[12] The expression for D in Eq. (2.24) is the directional derivative along the line from (a, b) to $(a+h, b+k)$ [see Eq. (2.19)] times the distance $\sqrt{h^2+k^2}$ between these two points.

[13] The numbers that multiply various terms in these expansions are the binomial coefficients, $\binom{n}{m} = \frac{n!}{(n-m)!m!}$. They appear in the expansion $(a+b)^n = \sum_{m=0}^{n} \binom{n}{m} a^{n-m} b^m$, and can be found from Pascal's triangle:

$$\begin{array}{ccccccccc}
 & & & & 1 & & & & \\
 & & & 1 & & 1 & & & \\
 & & 1 & & 2 & & 1 & & \\
 & 1 & & 3 & & 3 & & 1 & \\
1 & & 4 & & 6 & & 4 & & 1
\end{array}$$

2.8 Stationary points

Definition. The point (a, b) is a *stationary point* of the function $f(x, y)$ if

$$f'_x(a, b) = 0 \quad \text{and} \quad f'_y(a, b) = 0.$$

From Eq. (2.26), the Taylor expansion at the stationary point is

$$f(a + h, b + k) = f(a, b) + \frac{1}{2}\left[f''_{xx}h^2 + 2f''_{xy}hk + f''_{yy}k^2\right] + \dots . \tag{2.27}$$

The nature of the stationary point is determined by studying the second-order term (assuming that at least some of the second-order derivatives are nonzero). Let $A = f''_{xx}(a, b)$, $B = f''_{xy}(a, b)$, $C = f''_{yy}(a, b)$, and

$$\Delta = AC - B^2. \tag{2.28}$$

For $\Delta > 0$ the stationary point is an extremum; for $A > 0$ it is a minimum, for $A < 0$ it is a maximum. For $\Delta < 0$ the stationary point is a saddle point.

<u>Proof.</u>
The expression in square brackets in Eq. (2.27) can be written as

$$k^2\left(f''_{xx}\xi^2 + 2f''_{xy}\xi + f''_{yy}\right),$$

where $\xi = h/k$ (for $k \neq 0$). The quadratic function in brackets has no roots (i.e., does not vanish) if the discriminant

$$4{f''_{xy}}^2 - 4f''_{xx}f''_{yy} = -4\left(f''_{xx}f''_{yy} - {f''_{xy}}^2\right) = -4\Delta < 0, \text{ i.e., } \Delta > 0.$$

For $f''_{xx} > 0$ the expression is positive (minimum), for $f''_{xx} < 0$ it is negative (maximum). For $\Delta < 0$, the quadratic function does have roots for nonzero h and k, which means that we have a saddle point.

For $\Delta = 0$ further investigation is needed (i.e., higher-order terms in the Taylor expansion need to be examined). For example, $f(x, y) = x^4 + y^4$ has a stationary point $(0, 0)$, and this is a minimum, since $f(0, 0) = 0$ and $f(x, y) > 0$ for $(x, y) \neq (0, 0)$. However, this could not be concluded by studying the second-order derivatives, since $\Delta = 0$ for this function at $(0, 0)$.

3 Vector functions

3.1 Basic ideas

In Ch. 2 we considered functions of several variables, e.g., $f(x, y, z)$. Such a function can describe, e.g., the temperature in the room or outside. What about wind? Wind has speed (i.e., *magnitude*), but its *direction* also matters. In fact we are concerned with the wind *velocity vector* at a given point[14],

$$\boldsymbol{v}(x, y, z).$$

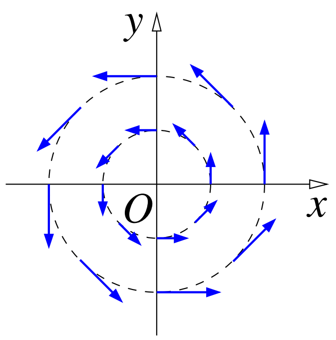
This is a *vector function*. It has three components, v_x, v_y and v_z at each point.

Vector functions are also called *vector fields*. Other examples of vector functions are an electric field of a distribution of electric charges, $\boldsymbol{E}(x, y, z)$, or a magnetic field $\boldsymbol{B}(x, y, z)$ created by a magnet or a current.

Example 1. If a liquid rotates as a whole in a cylindrical glass, its velocity increases linearly with the distance from the axis (z) and is perpendicular to the radius $\boldsymbol{\rho} = (x, y, 0)$ in the xy-plane:

$$\boldsymbol{v} = (-by, bx, 0), \tag{3.1}$$

where b is a constant.



Indeed, we can check that $\boldsymbol{v}$ is perpendicular to $\boldsymbol{\rho}$, since $\boldsymbol{v} \cdot \boldsymbol{\rho} = 0$. The magnitude of $\boldsymbol{v}$ is $|\boldsymbol{v}| = b\sqrt{x^2 + y^2} = b\rho$, where $\rho = |\boldsymbol{\rho}|$, i.e., the velocity increases linearly with ρ. Equation (3.1) can be written as

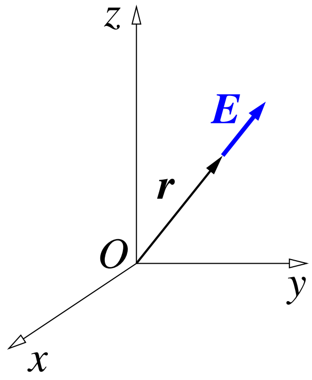
$$\boldsymbol{v} = \boldsymbol{b} \times \boldsymbol{r}, \tag{3.2}$$

where $\boldsymbol{r}$ is the position vector (1.8), $\boldsymbol{b} = b\boldsymbol{k}$ and $\boldsymbol{k}$ is the unit vector along z axes. Indeed,

$$\boldsymbol{v} = \boldsymbol{b} \times \boldsymbol{r} = \begin{vmatrix} \boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k} \\ 0 & 0 & b \\ x & y & z \end{vmatrix} = \boldsymbol{i} \begin{vmatrix} 0 & b \\ y & z \end{vmatrix} - \boldsymbol{j} \begin{vmatrix} 0 & b \\ x & z \end{vmatrix} + \boldsymbol{k} \begin{vmatrix} 0 & 0 \\ x & y \end{vmatrix} = -by\boldsymbol{i} + bx\boldsymbol{j},$$

which is the same as (3.1).

[14] If we consider the dependence of the velocity on time t, this will be a vector function of four variables $\boldsymbol{v}(x, y, z, t)$.



Example 2. The electric field of a positive charge placed at the origin is directed away from it and its magnitude is inversely proportional to the squared distance from it:

$$\boldsymbol{E} = \frac{k}{r^2}\frac{\boldsymbol{r}}{r} = \frac{k\boldsymbol{r}}{r^3}, \tag{3.3}$$

where k is a constant, and $\boldsymbol{r}/r$ is a unit vector along $\boldsymbol{r}$. In components,

$$\begin{aligned} \boldsymbol{E} &= \frac{k(x\boldsymbol{i} + y\boldsymbol{j} + z\boldsymbol{k})}{(x^2+y^2+z^2)^{3/2}} \\ &= \left(\frac{kx}{(x^2+y^2+z^2)^{3/2}}, \frac{ky}{(x^2+y^2+z^2)^{3/2}}, \frac{kz}{(x^2+y^2+z^2)^{3/2}}\right). \end{aligned} \tag{3.4}$$

As opposed to vector functions, $f(x, y, z)$ is sometimes called a *scalar function.*

3.2 Divergence and curl

In Sec. 2.6 we introduced the operator of "three-dimensional derivative" ∇, Eq. (2.17). When this vector operator acts on a scalar function f we obtain

$$\nabla f = \frac{\partial f}{\partial x}\boldsymbol{i} + \frac{\partial f}{\partial y}\boldsymbol{j} + \frac{\partial f}{\partial z}\boldsymbol{k},$$

which is the gradient vector field. The direction of ∇f is that in which f grows fastest, and its magnitude gives the corresponding rate of growth.

How can we apply the vector operator ∇ to a generic vector field $\boldsymbol{A}(x, y, x)$?

We know that two vectors $\boldsymbol{a}$ and $\boldsymbol{b}$ can be multiplied in two ways, i.e., as a scalar product,

$$\boldsymbol{a} \cdot \boldsymbol{b} = a_x b_x + a_y b_y + a_z b_z, \tag{3.5}$$

or as a vector product[15],

$$\boldsymbol{a} \times \boldsymbol{b} = \begin{vmatrix} \boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix} = (a_y b_z - a_z b_y)\boldsymbol{i} + (a_z b_x - a_x b_z)\boldsymbol{j} + (a_x b_y - a_y b_x)\boldsymbol{k}. \tag{3.6}$$

[15] Recall that the terms in the final expression for the cross product (3.6) are obtained from each other by cyclic permutations $\begin{array}{ccc} \boldsymbol{i} & \longrightarrow & \boldsymbol{j} \\ \nwarrow & \boldsymbol{k} & \swarrow \end{array}$ and $\begin{array}{ccc} x & \longrightarrow & y \\ \nwarrow & z & \swarrow \end{array}$.

Accordingly, we can construct two different "derivatives" of a vector field. The first,

$$\nabla \cdot \boldsymbol{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}, \tag{3.7}$$

is a scalar, and is called the *divergence* of $\boldsymbol{A}$ (sometimes denoted $\mathrm{div}\boldsymbol{A}$). The second,

$$\nabla \times \boldsymbol{A} = \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right)\boldsymbol{i} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)\boldsymbol{j} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)\boldsymbol{k}, \tag{3.8}$$

is a vector, and is called the *curl* of $\boldsymbol{A}$ (sometimes denoted $\mathrm{curl}\boldsymbol{A}$).

Examples.

1. Let $f(x, y, z) = r = \sqrt{x^2 + y^2 + z^2}$. Its gradient is

$$\nabla r = \frac{x}{\sqrt{x^2+y^2+z^2}}\boldsymbol{i} + \frac{y}{\sqrt{x^2+y^2+z^2}}\boldsymbol{j} + \frac{z}{\sqrt{x^2+y^2+z^2}}\boldsymbol{k} = \frac{\boldsymbol{r}}{r}, \tag{3.9}$$

   which is a unit vector in the direction of $\boldsymbol{r}$.

2. For $f = \boldsymbol{a} \cdot \boldsymbol{r}$, where $\boldsymbol{a}$ is a constant vector, the gradient is

$$\nabla(\boldsymbol{a} \cdot \boldsymbol{r}) = \nabla(a_x x + a_y y + a_z z) = a_x\boldsymbol{i} + a_y\boldsymbol{j} + a_z\boldsymbol{k} = \boldsymbol{a}. \tag{3.10}$$

3. For $\boldsymbol{A} = \boldsymbol{r}$, using (3.7), the divergence $\nabla \cdot \boldsymbol{A}$ is found as

$$\nabla \cdot \boldsymbol{r} = \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} = 3. \tag{3.11}$$

4. For $\boldsymbol{A} = \boldsymbol{r}$, using (3.8), the curl $\nabla \times \boldsymbol{A}$ is found as

$$\nabla \times \boldsymbol{r} = (0-0)\boldsymbol{i} + (0-0)\boldsymbol{j} + (0-0)\boldsymbol{k} = 0. \tag{3.12}$$

5. For $\boldsymbol{v} = (-by, bx, 0)$ from Eq. (3.1),

$$\nabla \times \boldsymbol{v} = 0\boldsymbol{i} + 0\boldsymbol{j} + (b - (-b))\boldsymbol{k} = 2b\boldsymbol{k} = 2\boldsymbol{b}. \tag{3.13}$$

   It is also easy to see that $\nabla \cdot \boldsymbol{v} = 0$.

   One can also obtain these answers using $\boldsymbol{v} = \boldsymbol{b} \times \boldsymbol{r}$ from Eq. (3.2). For example,

$$\nabla \cdot (\boldsymbol{b} \times \boldsymbol{r}) = -\boldsymbol{b} \cdot (\nabla \times \boldsymbol{r}) = 0,$$

   where we treated ∇ as a usual vector, and used properties of the triple scalar product[16] and Eq. (3.12). (When doing such manipulations it is important to keep

[16] The triple scalar product $\boldsymbol{a} \cdot (\boldsymbol{b} \times \boldsymbol{c})$ is invariant (i.e., remains unchanged) upon a cyclic permutation of the vectors $\boldsymbol{a}$, $\boldsymbol{b}$ and $\boldsymbol{c}$, $\begin{array}{c} \boldsymbol{a} \longrightarrow \boldsymbol{b} \\ \nwarrow \boldsymbol{c} \swarrow \end{array}$:

$$\boldsymbol{a} \cdot (\boldsymbol{b} \times \boldsymbol{c}) = \boldsymbol{b} \cdot (\boldsymbol{c} \times \boldsymbol{a}) = \boldsymbol{c} \cdot (\boldsymbol{a} \times \boldsymbol{b}),$$

but changes sign if any two vectors are swapped, e.g., $\boldsymbol{a} \cdot (\boldsymbol{b} \times \boldsymbol{c}) = -\boldsymbol{b} \cdot (\boldsymbol{a} \times \boldsymbol{c})$.

∇ to the left of the vector $\boldsymbol{r}$ on which it acts.) To find the curl, we use the "*bac minus cab*" rule for the triple vector product[17],

$$\begin{aligned}
\nabla \times (\boldsymbol{b} \times \boldsymbol{r}) &= \boldsymbol{b}(\nabla \cdot \boldsymbol{r}) - (\boldsymbol{b} \cdot \nabla)\boldsymbol{r} \\
&= 3\boldsymbol{b} - \left(b_x \frac{\partial}{\partial x} + b_y \frac{\partial}{\partial y} + b_z \frac{\partial}{\partial z}\right)(x\boldsymbol{i} + y\boldsymbol{j} + z\boldsymbol{k}) \\
&= 3\boldsymbol{b} - (b_x\boldsymbol{i} + b_y\boldsymbol{j} + b_z\boldsymbol{k}) \\
&= 2\boldsymbol{b},
\end{aligned}$$

exactly as in Eq. (3.13). Here we used Eq. (3.11) and wrote the second term in the top line so that $\boldsymbol{r}$ were kept to the right of ∇.

The "tricks" we used above to find $\nabla \cdot (\boldsymbol{b} \times \boldsymbol{r})$ and $\nabla \times (\boldsymbol{b} \times \boldsymbol{r})$ are particular cases of general vector identities involving ∇ and some scalar and vector functions.

3.3 Vector identities

The vector identities that involve ∇ generalise the basic product rule for functions of one variable, $(uv)' = u'v + uv'$.

There are several such identities, since the functions on which ∇ acts can be scalar of vector fields, and the way ∇ acts on vector fields can be either through a dot or through a cross product.

$$\nabla(fg) = \nabla f\, g + f\nabla g, \tag{3.14}$$
$$\nabla \cdot (f\boldsymbol{A}) = \nabla f \cdot \boldsymbol{A} + f\nabla \cdot \boldsymbol{A}, \tag{3.15}$$
$$\nabla \times (f\boldsymbol{A}) = \nabla f \times \boldsymbol{A} + f\nabla \times \boldsymbol{A}, \tag{3.16}$$
$$\nabla \cdot (\boldsymbol{A} \times \boldsymbol{B}) = \boldsymbol{B} \cdot (\nabla \times \boldsymbol{A}) - \boldsymbol{A} \cdot (\nabla \times \boldsymbol{B}), \tag{3.17}$$
$$\nabla \times (\boldsymbol{A} \times \boldsymbol{B}) = (\boldsymbol{B} \cdot \nabla)\boldsymbol{A} - \boldsymbol{B}(\nabla \cdot \boldsymbol{A}) + \boldsymbol{A}(\nabla \cdot \boldsymbol{B}) - (\boldsymbol{A} \cdot \nabla)\boldsymbol{B}. \tag{3.18}$$

To prove these identities, we use the definitions of the relevant vector operations and the usual product rule for partial derivatives. For example, expanding the left-hand side of Eq. (3.15), according to Eq. (3.7), we obtain:

$$\begin{aligned}
\nabla \cdot (f\boldsymbol{A}) &= \frac{\partial}{\partial x}(fA_x) + \frac{\partial}{\partial y}(fA_y) + \frac{\partial}{\partial z}(fA_z) \\
&= \frac{\partial f}{\partial x}A_x + f\frac{\partial A_x}{\partial x} + \frac{\partial f}{\partial y}A_y + f\frac{\partial A_y}{\partial y} + \frac{\partial f}{\partial z}A_z + f\frac{\partial A_z}{\partial z} \\
&= \left(\frac{\partial f}{\partial x}A_x + \frac{\partial f}{\partial y}A_y + \frac{\partial f}{\partial z}A_z\right) + f\left(\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}\right) \\
&= \nabla f \cdot \boldsymbol{A} + f\nabla \cdot \boldsymbol{A},
\end{aligned}$$

which is the right-hand side of Eq. (3.15).

There are also several identities involving second-order derivatives, i.e., two successive applications of the ∇ operators.

[17] The triple vector product satisfies the $bac - cab$ identity $\boldsymbol{a} \times (\boldsymbol{b} \times \boldsymbol{c}) = \boldsymbol{b}(\boldsymbol{a} \cdot \boldsymbol{c}) - \boldsymbol{c}(\boldsymbol{a} \cdot \boldsymbol{b})$.

Taking the divergence of the gradient of a scalar function, we find

$$\nabla \cdot \nabla f \equiv \nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}, \tag{3.19}$$

where the operator

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \tag{3.20}$$

is called the *Laplacian.*

The curl of the gradient of a scalar function is identically zero,

$$\nabla \times \nabla f = 0, \tag{3.21}$$

and so is the divergence of the curl of a vector field,

$$\nabla \cdot \nabla \times \boldsymbol{A} = 0. \tag{3.22}$$

Finally, the curl of the curl of a vector field is expressed in a $bac - cab$-like identity,

$$\nabla \times (\nabla \times \boldsymbol{A}) = \nabla(\nabla \cdot \boldsymbol{A}) - \nabla^2 \boldsymbol{A}. \tag{3.23}$$

To prove Eq. (3.21), apply the curl operator (3.8) to the components of the gradient vector,

$$\begin{aligned}
\nabla \times \nabla f = \nabla \times &\left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right) \\
&= \left(\frac{\partial}{\partial y}\frac{\partial f}{\partial z} - \frac{\partial}{\partial z}\frac{\partial f}{\partial y}, \frac{\partial}{\partial z}\frac{\partial f}{\partial x} - \frac{\partial}{\partial x}\frac{\partial f}{\partial z}, \frac{\partial}{\partial x}\frac{\partial f}{\partial y} - \frac{\partial}{\partial y}\frac{\partial f}{\partial x}\right) \\
&= \left(\frac{\partial^2 f}{\partial y \partial z} - \frac{\partial^2 f}{\partial z \partial y}, \frac{\partial^2 f}{\partial z \partial x} - \frac{\partial^2 f}{\partial x \partial z}, \frac{\partial^2 f}{\partial x \partial y} - \frac{\partial^2 f}{\partial y \partial x}\right) = (0, 0, 0) = 0.
\end{aligned}$$

Other identities are proved in a similar way.

3.4 Potential and solenoidal fields

Definition. A vector field $\boldsymbol{A}$ is called *potential* (or *irrotational*) if

$$\nabla \times \boldsymbol{A} = 0. \tag{3.24}$$

Comparison with the identity (3.21) suggests that an irrotational vector field $\boldsymbol{A}$ can be written as the gradient of a scalar function,

$$\boldsymbol{A} = \nabla U, \tag{3.25}$$

where the scalar function $U(x, y, z)$ is called the *potential* (see Sec. 4.2).

Definition. A vector field $\boldsymbol{B}$ is *solenoidal* if

$$\nabla \cdot \boldsymbol{B} = 0. \tag{3.26}$$

Comparison with the identity (3.22) indicates that a solenoidal vector field $\boldsymbol{B}$ can be written as the curl of vector field,

$$\boldsymbol{B} = \nabla \times \boldsymbol{A}, \tag{3.27}$$

with $\boldsymbol{A}$ known as the *vector potential*.

Note that neither the potential U defined by Eq. (3.25), nor the vector potential $\boldsymbol{A}$ defined by Eq. (3.27), is unique. Thus, using a function $U + C$, where C is a constant, instead of U in Eq. (3.25), does not change its validity (since $\nabla C = 0$). Similarly, replacing $\boldsymbol{A}$ by $\boldsymbol{A} + \nabla\chi$, where χ is any scalar function, in Eq. (3.27), does not affect its validity, because $\nabla \times \nabla\chi = 0$.

4 Line integrals and double integrals

In this chapter we extend the theory of integration to curves and surfaces in space. The resulting theory of **line** and **surface** integrals gives powerful mathematical tools for applications in science and engineering. For example, line integrals define the work done by a force in moving an object along a path, whereas a surface integral can be used to find the rate of flow of a fluid across a surface. We begin by describing two methods of integrating along a curve in space, scalar line integrals (first kind) and vector line integrals (second kind).

4.1 Line integral of the first kind

We first consider line integrals of scalar-valued functions over curves in the plane. Consider a curve in space with length element ds. In Sec. 1.2 we introduced an integral that gives the arc length of the curve, Eq. (1.19). More generally, one can consider integrals of the form

$$\int\limits_C f(x,y,z)ds, \tag{4.1}$$

where $f(x,y,z)$ is some function taken at the points (x,y,z) on the curve, and $\int\limits_C$ means that the integral is taken *along the curve* C.

The integral (4.1) can be thought of as the "mass of the curve", where f represents the linear mass density (e.g., mass per unit length) of the curve.

If the curve is given parametrically, as in Eq. (1.9), the integral takes the form

$$\int_a^b f(x(t),y(t),z(t))\sqrt{[x'(t)]^2+[y'(t)]^2+[z'(t)]^2}\,dt, \tag{4.2}$$

where a and b correspond to the initial and final points $\boldsymbol{r}(a)$ and $\boldsymbol{r}(b)$ on C.

Example 1. Integrate the function $f(x,y,z)=e^{-\alpha z}$ over one turn of the helix,

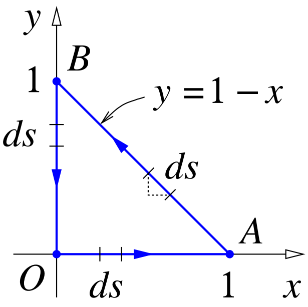
$$x=a\cos t,\quad y=a\sin t,\quad z=bt,$$

between points $t=0$ and $t=2\pi$.

The length element for the helix is $ds=\sqrt{a^2+b^2}dt$ (Sec. 1.2), so we have

$$\int_C e^{-\alpha z}ds=\int_0^{2\pi} e^{-\alpha bt}\sqrt{a^2+b^2}dt=\frac{\sqrt{a^2+b^2}}{\alpha b}\left[1-e^{-2\pi\alpha b}\right].$$

Example 2. Find $\int\limits_C(x+y)ds$, where C is a closed triangular path $OABO$ in the xy-plane, with $O(0,0)$, $A(1,0)$, $B(0,1)$.



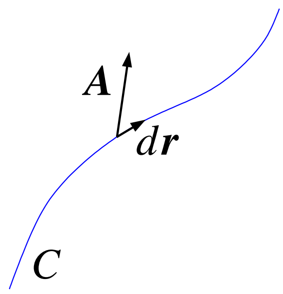
On OA: $y = 0$, $x + y = x$, $ds = dx$; on AB: $y = 1 - x$, $x + y = 1$, $ds = -\sqrt{2}dx$ (since x decreases, $dx < 0$, but we must always have $ds > 0$)[18]; on BO: $x = 0$, $x + y = y$, $ds = -dy$ (y decreases). Hence,

$$\int\limits_C (x + y)ds = \int_0^1 x dx + \int_1^0 \sqrt{2}(-dx) + \int_1^0 y(-dy) = 1 + \sqrt{2}.$$

Note that the line integral of the first kind does not depend on the direction in which C is traversed, as $ds > 0$ is always assumed [hence, $a < b$ in Eq. (4.2)].

4.2 Line integral of the second kind

We are now going to see how to integrate a vector field along a curve in space.



For vector functions, we can consider integrals of the type

$$\int\limits_C \boldsymbol{A} \cdot d\boldsymbol{r}. \tag{4.3}$$

Be clear about the notation used here. In the vector line integral 4.3 the differential term $d\boldsymbol{r}$ should be thought of as a vector quantity (namely, the "differential" of position along the path, whereas in the scalar line integral 4.1 the differential term ds is a scalar quantity (namely, the differential of arc length). Also remember that the $\cdot$ term in 4.3 represents vector multiplication (dot product).

[18] Formally, the element of length is $ds = \sqrt{dx^2 + dy^2}$ [cf. Eq. (1.18)]. On AB we have $y = 1 - x$, so that $dy = -dx$ and $ds = \sqrt{2dx^2} = \sqrt{2}|dx| = -\sqrt{2}dx$ (since $dx < 0$ on AB).

In Cartesian coordinates, $d\boldsymbol{r} = dx\boldsymbol{i} + dy\boldsymbol{j} + dz\boldsymbol{k}$, and the integral takes the form

$$\int\limits_C (A_x dx + A_y dy + A_z dz), \tag{4.4}$$

also written $\int\limits_C (Pdx + Qdy + Rdz)$, where P, Q and R are functions of x, y, z.

The integral (4.3) does depend on the direction in which C is traversed, i.e.,

$$\int_A^B \boldsymbol{A} \cdot d\boldsymbol{r} = -\int_B^A \boldsymbol{A} \cdot d\boldsymbol{r}, \tag{4.5}$$

where the integrals are taken along the same curve between points A and B, or in the opposite direction, from B to point A. (The change in direction changes $d\boldsymbol{r}$ into $-d\boldsymbol{r}$.)

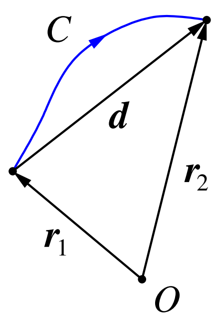
Example. Of this type is *work* by a force $\boldsymbol{F}$ in mechanics,

$$W = \int\limits_C \boldsymbol{F} \cdot d\boldsymbol{r}.$$

Let us find work by a *constant* force $\boldsymbol{F}$ along some curve C that connects points $\boldsymbol{r}_1$ and $\boldsymbol{r}_2$:

$$W = \int_{\boldsymbol{r}_1}^{\boldsymbol{r}_2} \boldsymbol{F} \cdot d\boldsymbol{r} = \boldsymbol{F} \cdot \int_{\boldsymbol{r}_1}^{\boldsymbol{r}_2} d\boldsymbol{r} = \boldsymbol{F} \cdot [\boldsymbol{r}]_{\boldsymbol{r}_1}^{\boldsymbol{r}_2} = \boldsymbol{F} \cdot (\boldsymbol{r}_2 - \boldsymbol{r}_1) = \boldsymbol{F} \cdot \boldsymbol{d},$$

where $\boldsymbol{d} = \boldsymbol{r}_2 - \boldsymbol{r}_1$ is the displacement vector.



We see that in this case the integral does not depend on the shape of the path, but only on the initial and final points. In fact, this is true for any vector function of the form $\boldsymbol{A} = \nabla U$. Indeed, for a path C that connects points $\boldsymbol{r}_1$ and $\boldsymbol{r}_2$, we have

$$\int\limits_C \nabla U \cdot d\boldsymbol{r} = \int_{\boldsymbol{r}_1}^{\boldsymbol{r}_2} dU = [U(\boldsymbol{r})]_{\boldsymbol{r}_1}^{\boldsymbol{r}_2} = U(\boldsymbol{r}_2) - U(\boldsymbol{r}_1), \tag{4.6}$$

where in the first step we used an expression for the total differential (Sec. 2.5),

$$dU = \frac{\partial U}{\partial x}dx + \frac{\partial U}{\partial y}dx + \frac{\partial U}{\partial z}dz = \nabla U \cdot d\boldsymbol{r}.$$

This allows us to find the *potential* U for any irrotational vector field $\boldsymbol{A}$ (see Sec. 3.4) as

$$U(\boldsymbol{r}) = \int_{\boldsymbol{r}_0}^{\boldsymbol{r}} \boldsymbol{A} \cdot d\boldsymbol{r}, \tag{4.7}$$

where $\boldsymbol{r}_0$ is an arbitrary point (e.g., the origin), and the integral is taken along *any* path connecting $\boldsymbol{r}_0$ with $\boldsymbol{r}$ (see Sec. 6.4).

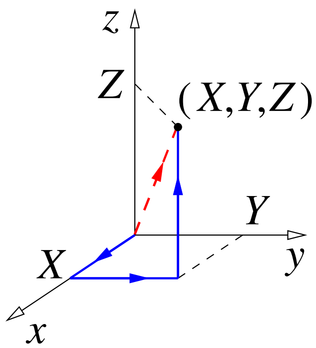
The potential defined by Eq. (4.7) depends on the choice of $\boldsymbol{r}_0$. However, a change in $\boldsymbol{r}_0$ amounts to adding a constant to U, which does not affect (3.25).

<u>Example.</u> Verify that $\boldsymbol{A} = yz\boldsymbol{i} + zx\boldsymbol{j} + xy\boldsymbol{k}$ is irrotational and find the potential.

To verify that $\boldsymbol{A}$ is irrotational we calculate

$$\begin{aligned}\nabla \times \boldsymbol{A} &= \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right)\boldsymbol{i} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)\boldsymbol{j} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)\boldsymbol{k} \\ &= (x-x)\boldsymbol{i} + (y-y)\boldsymbol{j} + (z-z)\boldsymbol{k} = 0.\end{aligned}$$

To find the potential at point (X, Y, Z) we choose C that starts at the origin and consists



of three straight segments parallel to the coordinate axes, from $(0, 0, 0)$ to $(X, 0, 0)$, then to $(X, Y, 0)$, and finally to (X, Y, Z). We then have

$$\begin{aligned}U(X,Y,Z) &= \int\limits_C \boldsymbol{A} \cdot d\boldsymbol{r} = \int\limits_C (yzdx + zxdy + xydz) \\ &= \int\limits_{(0,0,0)}^{(X,0,0)} yzdx + \int\limits_{(X,0,0)}^{(X,Y,0)} zxdy + \int\limits_{(X,Y,0)}^{(X,Y,Z)} xydz = 0 + 0 + XYZ = XYZ.\end{aligned}$$

Hence the potential is

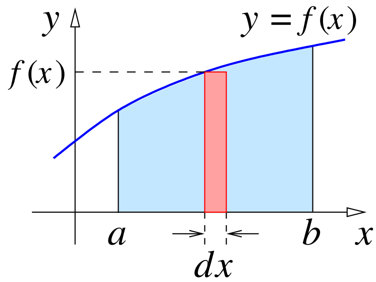
$$U = xyz. \tag{4.8}$$

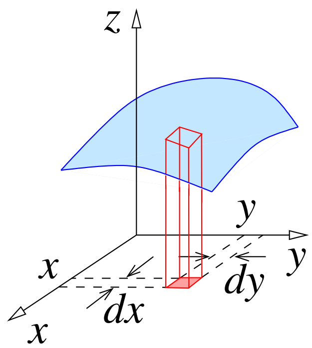
It is easy to verify that $\nabla(xyz) = \boldsymbol{A}$, as required. Note that $\nabla U = \boldsymbol{A}$ will also hold if we add any constant to the right-hand side of Eq. (4.8).

<u>Exercise.</u> Verify that the same potential (4.8) is obtained if C is chosen as a straight line segment between $(0, 0, 0)$ and (X, Y, Z). [Hint: this path can be parameterised by $x = Xt$, $y = Yt$, $z = Zt$, $0 \le t \le 1$.]

4.3 Double integrals

In Ch. 2 we introduced functions of several variables and learned how to differentiate then using partial differentiation. The next step is to carry out a similar process with respect to integration. As you know, single or one-variable integrals are developed from Riemann sums and they are used to compute *areas* of regions in $\mathbf{R}^2$. In a similar way, we use Riemann sums to develop double or two-variable and triple or three-variable integrals, which are used to compute *volumes* of solid regions in $\mathbf{R}^3$. These multiple integrals have many applications in statistics, science and engineering.





Recall the meaning of the definite integral

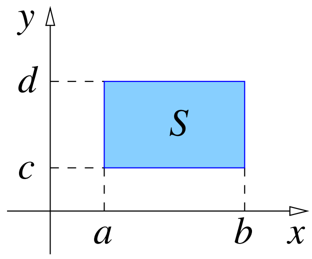
$$\int_a^b f(x)dx.$$

It gives the area in the xy-plane under curve $y = f(x)$. Here $f(x)dx$ is the area of a narrow strip of width dx and height $f(x)$, and the integral is just the *sum* of such contributions (hence the symbol $\int$ that resembles letter S).

Consider now a function of two variables $z = f(x, y)$ which describes a surface in three-dimensional space. If we consider small increments dx and dy, then the points in the xy-plane with coordinates in the intervals $[x, x + dx]$ and $y, y + dy]$ is a small rectangle. Its area is $dxdy$, so that $dS = dxdy$ is the *element of area* in the xy-plane.

The quantity $f(x, y)dxdy$ is the volume of a narrow rectangular "column" that stretches between the xy-plane and the surface. If we now *add* all such columns for x and y in a certain region, we will find the *volume under the surface*. This is written as

$$V = \iint\limits_S f(x, y)dxdy, \tag{4.9}$$



which is a *double integral* over the region S.

It is easiest to calculate double integrals when S is a rectangle with sides parallel to the x and y axes, e.g., $a \le x \le b$, $c \le y \le d$, so that

$$\iint\limits_S f(x,y)dxdy = \underbrace{\int_c^d \underbrace{\int_a^b f(x,y)dx}_{\text{with respect to } x} dy}_{\text{with respect to } y}.$$

This can also be written as an *iterated* integral,

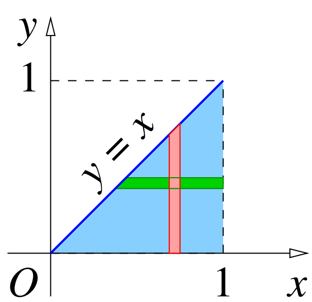
$$\int_c^d dy \int_a^b f(x,y)dx, \tag{4.10}$$

where the integral on the right (over x) is to be done first[19].

Example 1. Find the volume under the surface $f(x,y) = x + 2y$ for $0 \le x \le 2$, $1 \le y \le 3$ ("a house with a mono-pitched roof").

$$\begin{aligned}\int_1^3 dy \int_0^2 (x+2y)dx &= \int_1^3 dy \left[\frac{x^2}{2} + 2yx\right]_0^2 \\ &= \int_1^3 dy(2+4y) = \left[2y + \frac{4y^2}{2}\right]_1^3 = 20.\end{aligned}$$

Exercise. Verify that $\int_0^2 dx \int_1^3 (x+2y)dy$ gives the same result.



[19] One can also write (4.10) as $\int_a^b dx \int_c^d f(x,y)dy$. For continuous or piecewise continuous functions, both integrals give the same answer (because they describe the same volume).

Example 2. For a triangular region S: $0 \le x \le 1$, $0 \le y \le x$, the double integral can be written as the following iterated integral:

$$\iint\limits_S f(x,y)dxdy = \int_0^1 dx \int_0^x f(x,y)dy. \tag{4.11}$$

One can also *change the order of integration* by representing S as $0 \le y \le 1$, $y \le x \le 1$, and write the same integral as

$$\int_0^1 dy \int_y^1 f(x,y)dx. \tag{4.12}$$

Exercise. Verify that (4.11) and (4.12) give the same answer for some $f(x,y)$, e.g., $f(x,y) = x + 2y$, $f(x,y) = xy$, $f(x,y) = x^2 + y^2$, $f(x,y) = \dfrac{1}{x+y}$, etc.[20]

Example 3. Let S be a disc of radius a: $x^2 + y^2 \le a^2$, and $z = \sqrt{a^2 - x^2 - y^2}$, describing a spherical "dome". Find the volume of the dome (i.e., hemisphere).

For the points in the disc, $-a \le x \le a$ and $-\sqrt{a^2 - x^2} \le y \le \sqrt{a^2 - x^2}$, so the volume is

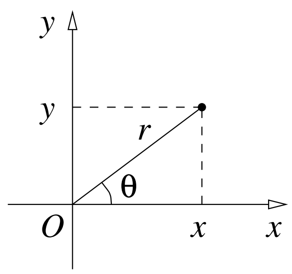
$$V = \int\limits_{-a}^{a} dx \int\limits_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} \sqrt{a^2 - x^2 - y^2}\, dy. \tag{4.13}$$

This is not the simplest integral to evaluate in Cartesian coordinates!

4.4 Polar coordinates

Sometimes when evaluating these double integrals it is much easier to adopt a co-ordinate system which is more suited to the geometry of the region of integration. In 2D we know two co-ordinate systems, the cartesian (x, y) system or the polar (r, θ) co-ordinate system. These polar coordinate transformations can prove to be especially convenient when dealing with regions whose boundaries are circles or parts of circles.

The polar coordinates in the plane are the *radius* r (distance from the origin) and *angle* θ, which is measured from the positive direction of the x-axis and is regarded positive when the rotation is anticlockwise.



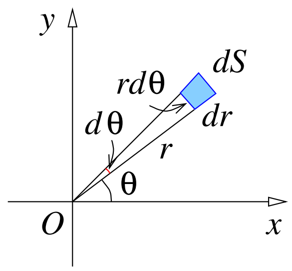
[20] It is interesting that for $f(x,y) = 1/(x+y)$ one of the integrals is much quicker to evaluate than the other (though, of course, both give the same answer, $\ln 2$).

In terms of r and θ, the Cartesian coordinates are

$$x = r\cos\theta, \quad y = r\sin\theta, \tag{4.14}$$

which gives[21]

$$r = \sqrt{x^2+y^2}, \quad \tan\theta = y/x. \tag{4.15}$$



Let us consider a point with coordinates (r, θ) and increase r by dr and θ by $d\theta$. The point will move through dr in the radial direction and through $rd\theta$ in the perpendicular direction[22], i.e., along the circle. Hence, the element of area in polar coordinates is

$$dS = rdrd\theta, \tag{4.16}$$

and the element of length is

$$ds = \sqrt{dr^2 + r^2 d\theta^2}. \tag{4.17}$$

We can now evaluate the integral in Example 3 of Sec. 4.3 using polar coordinates, in which $z = \sqrt{a^2-r^2}$, and the disc is described by $0 \le r \le a$, $0 \le \theta \le 2\pi$. Hence,

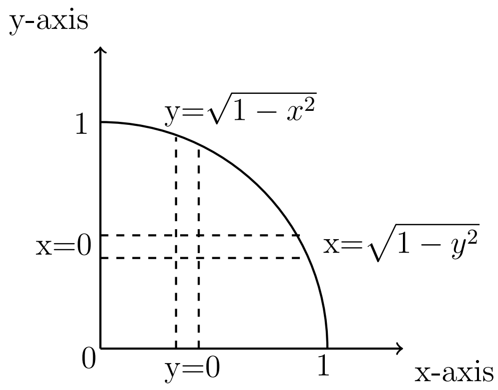
$$\begin{aligned} V &= \iint\limits_S \sqrt{a^2-x^2-y^2}\, rdrd\theta = \int_0^{2\pi} d\theta \int_0^a \sqrt{a^2-r^2}\, rdr \\ &= [\theta]_0^{2\pi}\left(-\frac{1}{2}\right)\int_0^a \sqrt{a^2-r^2}\, d(a^2-r^2) = -\pi\left[\frac{2}{3}(a^2-r^2)^{3/2}\right]_0^a = \frac{2\pi}{3}a^3, \end{aligned}$$

which is, as expected, one-half of the volume of the sphere, $\frac{4\pi}{3}a^3$.

Let's look at another example. Integrate x^2+y^2 over a quadrant of a unit circle.

[21] When finding θ from Eq. (4.15), use $\theta = \arctan(y/x)$ for $x > 0$, and $\theta = \arctan(y/x) + \pi$ (or $-\pi$) for $x < 0$.

[22] Recall that for a circle of radius R, the length of an arc that subtends angle α is $l = \alpha R$. Note also that for small dr and $rd\theta$, the two displacements form a small rectangle.



$$
\begin{aligned}
I &= \int_0^1 \int_0^{\sqrt{1-y^2}} (x^2+y^2)\, dxdy = \int_0^1 \Big(\frac{x^3}{3} + xy^2\Big)_0^{\sqrt{1-y^2}} dy \\
&= \int_0^1 \Big(\frac{(1-y^2)(1-y^2)^{1/2}}{3} + y^2\sqrt{1-y^2}\Big)\, dy \\
&= \int_0^1 \Big(\frac{1}{3}(1-y^2) + y^2\Big)\sqrt{1-y^2}\, dy = \int_0^1 \sqrt{1-y^2}\Big(\frac{1}{3} - \frac{y^2}{3} + y^2\Big)\, dy \\
&= \int_0^1 \sqrt{1-y^2}\Big(\frac{1}{3} + \frac{2y^2}{3}\Big)\, dy = \frac{1}{3}\int_0^1 \sqrt{1-y^2}\Big(1 + 2y^2\Big)\, dy
\end{aligned}
$$

To perform this integral we need to use the substitution

$$
y = \sin\theta \implies dy = \cos\theta d\theta
$$
$$
\sqrt{1-y^2} = \sqrt{1-\sin^2\theta} = \sqrt{\cos^2\theta} = \cos\theta
$$

Changing the limits we have

$$
\begin{aligned}
y &= 0 \implies \sin\theta = 0 \implies \theta = 0 \\
y &= 1 \implies \sin\theta = 1 \implies \theta = \pi/2
\end{aligned}
$$

and hence

$$
I = \int_0^{\pi/2} \frac{1}{3}\cos\theta(1 + 2\sin^2\theta)\cos\theta\, d\theta = \frac{1}{3}\int_0^{\pi/2} \cos^2\theta(1 + 2\sin^2\theta)\, d\theta
$$

Now $\sin 2\theta = 2\cos\theta\sin\theta$ and $\sin^2 2\theta = 4\cos^2\theta\sin^2\theta$ so that

$$
I = \frac{1}{3}\int_0^{\pi/2} (\cos^2\theta + \frac{1}{2}\sin^2 2\theta)\, d\theta = \frac{1}{3}\int_0^{\pi/2} \Big[\frac{1}{2}(1+\cos 2\theta) + \frac{1}{4}(1 - \cos 4\theta)\Big]\, d\theta
$$

Since $\cos^2\theta = \frac{1}{2}(1+\cos 2\theta)$ and $\sin^2\theta = \frac{1}{2}(1-\cos 2\theta)$

$$\begin{aligned}
I &= \frac{1}{3}\Big[\frac{1}{2}(\theta + \frac{1}{2}\sin 2\theta) + \frac{1}{4}(\theta - \frac{1}{4}\sin 4\theta\Big]_0^{\pi/2} \\
&= \frac{1}{3}\Big[\frac{1}{2}(\frac{\pi}{2} + \frac{1}{2}\sin\pi + \frac{1}{4}(\frac{\pi}{2} - \frac{1}{4}\sin 2\pi) - \frac{1}{2}(0+\sin 0) - \frac{1}{4}(0-\sin 0)\Big] \\
&= \frac{1}{3}\Big[\frac{\pi}{4} + \frac{\pi}{8}\Big] = \frac{\pi}{8}
\end{aligned}$$

What a long drawn out process that was! This can be simplified considerably by using a different co-ordinate system. If we switch to polar co-ordinates

$$x = r\cos\theta \qquad y = r\sin\theta \qquad dxdy = rdrd\theta \qquad x^2+y^2 = r^2$$

and the region of integration is

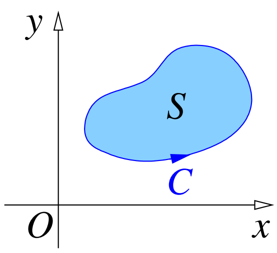
$$0 \le r \le 1 \qquad 0 \le \theta \le \pi/2$$

Then

$$I = \int_0^{\pi/2}\int_0^1 r^2\, rdrd\theta = \int_0^{\pi/2}\int_0^1 r^3\, drd\theta = \int_0^{\pi/2}\frac{1}{4}\,d\theta = \Big[\frac{\theta}{4}\Big]_0^{\pi/2} = \frac{\pi}{8}$$

much easier!

4.5 Green's theorem

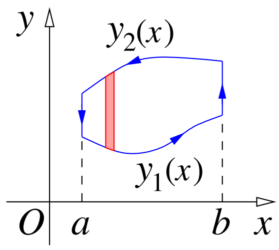


Green's theorem links a line integral of the second kind over a closed curve C in the xy-plane with a double integral over the region S enclosed by this curve:

$$\oint_C (Pdx + Qdy) = \iint_S \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dxdy. \tag{4.18}$$

Here $P(x,y)$ and $Q(x,y)$ are some functions, $\oint$ means that this integral is over a *closed* curve C, and the integral over C is taken in the anticlockwise direction (so that S remains on the left as C is traversed)[23].

[23] There are other formulae that link the integral of the derivative of a function with its values at the boundary, e.g., $\int_a^b \frac{df}{dx}dx = f(b) - f(a)$, or $\int_{\boldsymbol{r}_1}^{\boldsymbol{r}_2} \nabla U \cdot d\boldsymbol{r} = U(\boldsymbol{r}_2) - U(\boldsymbol{r}_1)$, where the integral is taken along any curve between points $\boldsymbol{r}_1$ and $\boldsymbol{r}_2$ [see Eq. (4.6)]. Two more such relations will be presented in Secs. 6.3 and 6.4.



Proof. Let us consider the case when the integration region S is $a \le x \le b$ and $y_1(x) \le y \le y_2(x)$, and $Q = 0$. Starting from the right-hand side of (4.18),

$$\begin{aligned}
-\iint\limits_S \frac{\partial P}{\partial y} dxdy &= -\int\limits_a^b dx \int\limits_{y_1(x)}^{y_2(x)} \frac{\partial P}{\partial y} dy = -\int_a^b dx \big[P(x,y)\big]_{y_1(x)}^{y_2(x)} \\
&= \int_a^b P(x, y_1(x))dx - \int_a^b P(x, y_2(x))dx \\
&= \int_a^b P(x, y_1(x))dx + \int_b^a P(x, y_2(x))dx = \oint\limits_C Pdx,
\end{aligned}$$

where C consists of two lines $x = a$ and $x = b$ (which do not contribute to the line integral), and two segments of curve, $y = y_1(x)$ traversed in the positive x direction, and $y = y_2(x)$ – in the negative x direction.

Since the P-parts and Q-parts are simply added on both sides of Eq. (4.18), a similar proof can be provided to show that the Q-parts are also equal.

Example. Green's formula can be used for finding areas by means of line integrals. For $P = -\frac{1}{2}y$ and $Q = \frac{1}{2}x$, the right-hand side of Eq. (4.18) gives

$$\iint\limits_S dxdy,$$

which is the area of region S. Hence, the area A of a region bounded by curve C is

$$A = \frac{1}{2} \oint\limits_C (-ydx + xdy). \tag{4.19}$$

Thus, the area of an ellipse: $x = a\cos t$, $y = b \sin t$, $0 \le t \le 2\pi$ (Sec. 5.1), is

$$\begin{aligned}
A &= \frac{1}{2} \int_0^{2\pi} [-b\sin t(-a \sin t dt) + a\cos t(b \cos t dt)] \\
&= \frac{1}{2} ab \int_0^{2\pi} (\sin^2 t + \cos^2 t) dt = \pi ab.
\end{aligned}$$

5 Surfaces and curvilinear coordinates

5.1 Equations of surfaces

An equation of the form $f(x, y, z) = 0$ or $f(x, y, z) = C$, where C is a constant, represents a surface.

Examples:

1. $ax + by + cz = d$ is the equation of a plane.

   In vector form, $(\boldsymbol{r} - \boldsymbol{a}) \cdot \boldsymbol{n} = 0$ or $\boldsymbol{r} \cdot \boldsymbol{n} = \boldsymbol{a} \cdot \boldsymbol{n}$, where $\boldsymbol{n}$ is a vector normal (i.e., perpendicular) to the plane, and $\boldsymbol{a}$ is a point in the plane.

2. $x^2 + y^2 + z^2 = a^2$ is the equation of a sphere of radius a, centred at the origin.

Alternatively, surfaces can be defined by equations in parametric form,

$$x = x(u, v), \quad y = y(u, v), \quad z = z(u, v), \tag{5.1}$$

or

$$\boldsymbol{r} = x(u, v)\boldsymbol{i} + y(u, v)\boldsymbol{j} + z(u, v)\boldsymbol{k}, \tag{5.2}$$

where u and v are *two* parameters. (Compare with the parametric equation of a curve (1.9) that contained only one parameter.)

Examples:

1. For $c \neq 0$, the plane from Example 1 above can be parameterised by

$$x = u, \quad y = v, \quad z = (d - au - bv)/c \quad (u \in \mathbb{R},\ v \in \mathbb{R}). \tag{5.3}$$

2. The sphere from Example 2 can be parameterised by

$$x = a \sin\theta \cos\phi, \quad y = a \sin\theta \sin\phi, \quad z = a \cos\theta, \tag{5.4}$$

   where the parameters θ and ϕ are similar to the latitude and longitude, with values in the range $0 \leq \theta \leq \pi$, $0 \leq \phi \leq 2\pi$.

   Verify that x, y and z from (5.4) satisfy $x^2 + y^2 + z^2 = a^2$.

For a surface in parametric form, $\boldsymbol{r} = \boldsymbol{r}(u, v)$, the two vectors, $\dfrac{\partial \boldsymbol{r}}{\partial u}$ and $\dfrac{\partial \boldsymbol{r}}{\partial v}$, are tangential to the surface. Their vector product,

$$\boldsymbol{n} = \frac{\partial \boldsymbol{r}}{\partial u} \times \frac{\partial \boldsymbol{r}}{\partial v}, \tag{5.5}$$

is a vector *normal* to the surface. If necessary, a unit normal can be constructed as $\boldsymbol{n}/|\boldsymbol{n}|$.

The equation of a plane *tangent* to the surface at point $\boldsymbol{r}_0 = \boldsymbol{r}(u_0, v_0)$ is

$$(\boldsymbol{r} - \boldsymbol{r}_0) \cdot \boldsymbol{n} = 0, \tag{5.6}$$

where the normal $\boldsymbol{n}$ is taken at (u_0, v_0).

For the surface given in the form $f(x, y, z) = C$, the normal can be found as

$$\boldsymbol{n} = \nabla f, \tag{5.7}$$

and the unit normal as $\nabla f/|\nabla f|$. Indeed, we know that the direction of ∇f is that in which f increases fastest, and the derivative of f in the direction of a unit vector $\boldsymbol{T}$ is given by $\nabla f \cdot \boldsymbol{T}$ (Sec. 2.6). Since f remains constant on the surface, $\nabla f \cdot \boldsymbol{T} = 0$ for *any* vector $\boldsymbol{T}$ tangential to the surface. Hence, the vector $\boldsymbol{n}$ from Eq. (5.7) is perpendicular to the surface, as required.

Example. For the plane $ax + by + cz = d$, the normal is found as

$$\boldsymbol{n} = \nabla(ax + by + cz) = a\boldsymbol{i} + b\boldsymbol{j} + c\boldsymbol{k},$$

and the unit normal is

$$\frac{a\boldsymbol{i} + b\boldsymbol{j} + c\boldsymbol{k}}{\sqrt{a^2 + b^2 + c^2}}.$$

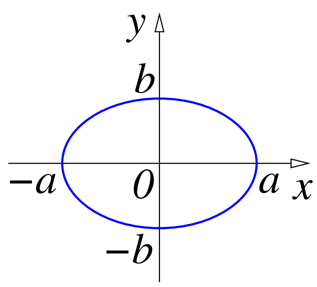
Exercise. For the plane in parametric form (5.3), find the normal using (5.5).

Second degree curves.

Before looking at the surfaces known as quadrics, let us explore the plane curves defined by the second-degree polynomial in x and y:

$$Ax^2 + By^2 + Cxy + Dx + Ey + F = 0. \tag{5.8}$$

By shifting the origin along the x and y axes and rotating the axes about the origin, one can eliminate the linear terms and the xy term from Eq. (5.8). We can then divide the equation by F to make the free term equal to unity (for $F \neq 0$). This leaves us with just three possible varieties of second-degree curves which have the following standard forms.



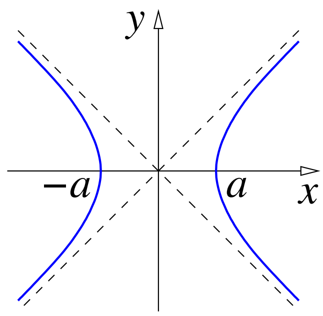
1. $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ is an *ellipse*.

   For $a > b$, a is the called semimajor axis, and b – the semiminor axis; for $b = a$ this is a circle. The ellipse can be parameterised by

   $$x = a\cos t, \quad y = b\sin t \qquad (0 \leq t \leq 2\pi). \tag{5.9}$$

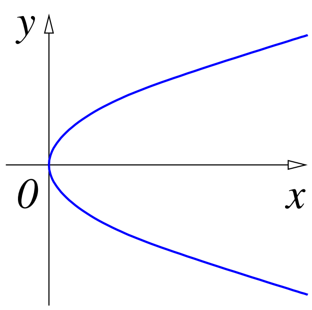
   Verify this by substitution into the Cartesian equation above.

2. $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$ is a *hyperbola*.



The hyperbola has two branches, one for $x > 0$ and one for $x < 0$, and two asymptotes $y = \pm bx/a$, which it approaches for large x and y.

3. $y^2 = 4ax$ is a *parabola*. (This case occurs when the coefficient of one of the quadratic terms vanishes.)



Exercise: show that the parabola is the locus of points such that the distance to point $(a, 0)$ (focus) equals the distance to the line $x = -a$ (directrix), i.e., $\sqrt{(x-a)^2 + y^2} = x + a$.

These three curves are known collectively as conic sections or *conics*, as they are obtained as intersections of the surface of a cone with a plane (see Fig. 1).

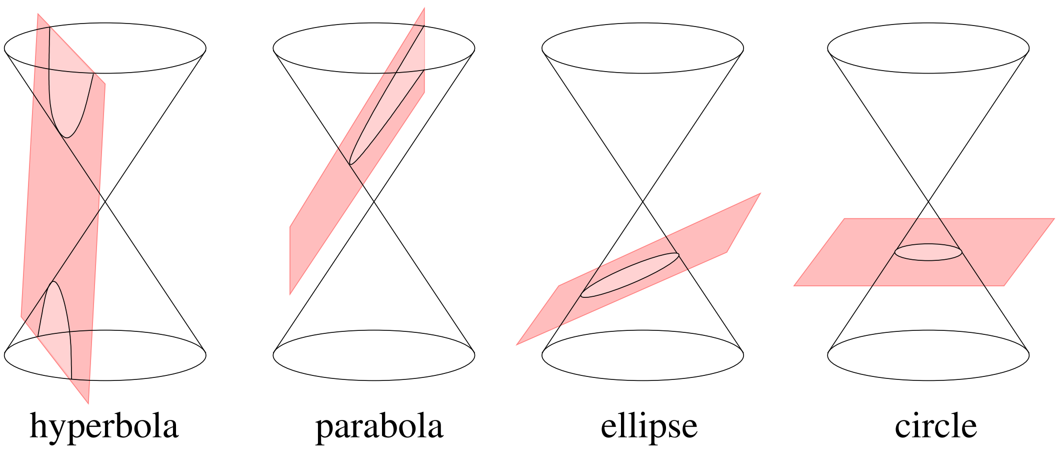


Figure 1: Conic sections.

In polar coordinates they are in fact descibed by the same equation,

$$r = \frac{p}{1 + e\cos\theta}. \tag{5.10}$$

Here $p > 0$ determines the size of the curve, and $e \geq 0$ (the *eccentricity*) determines its shape. For $e < 1$ the curve is an ellipse ($e = 0$ gives a circle), for $e = 1$ the curve is a parabola, and for $e > 1$ the curve is a hyperbola.

As we move from 2D to 3D we must first introduce the concept of a triple integral. Let $f(x, y, z))$ be a function of three variables. Analogous to the double integral, we define the triple integral of f over a solid region in space to be the limit of appropriate Riemann sums. If we use Cartesian coordinates, $dV = dxdydz$, this integral becomes a triple integral of the form

$$\iiint\limits_V f(x, y, z)dxdydz. \tag{5.11}$$

This is also called a *volume integral* and we will be looking more closely at integrals of this type later on in Chapter 6. To evaluate such integrals, we need to know the function f and the shape of the body, which determines the integration limits.

Example. Evaluate $\displaystyle\iiint\limits_V (x^2e^y + xyz)\,dV$, where V is the volume bounded by the closed cuboid whose faces are parallel to the coordinate planes so that

$$-2 \leq x \leq 3, \quad 0 \leq y \leq 1, \quad 0 \leq z \leq 5$$

The triple integral can be written as

$$\int\limits_{-2}^{3}\int\limits_{0}^{1}\int\limits_{0}^{5}(x^2e^y + xyz)dzdydx = \int\limits_{-2}^{3} dx \int\limits_{0}^{1} dy \int\limits_{0}^{5}(x^2e^y + xyz)dz$$

$$\int\limits_{-2}^{3} dx \int\limits_{0}^{1} \left(x^2e^yz + \frac{1}{2}xyz^2\right)_{z=0}^{z=5} dy = \int\limits_{-2}^{3} dx \int\limits_{0}^{1}(5x^2e^y + \frac{25}{2}xy)dy$$

$$\int\limits_{-2}^{3}\left(5x^2e^y + \frac{25}{4}xy^2\right)_{y=0}^{y=1} dx = \int\limits_{-2}^{3}(5(e-1)x^2 + \frac{25}{4}x)dx$$

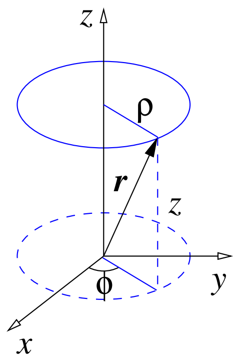
$$\left(\frac{5}{3}(e-1)x^3 + \frac{25}{8}x^2\right)_{x=-2}^{x=3} = \frac{175}{3}(e-1) + \frac{125}{8}$$

You can check that as f is a continuous function, and hence satisfies Fubini's theorem, that integrating in any of the five possible orders produces the same result.

Similar to double integrals, changes in the coordinate system from cartesian to polar can be used to transform a difficult integral to one which is relatively straightforward. In 3D we will concentrate on two such polar coordinate systems, cylindrical when integrating over solid bodies possessing an axis of rotational symmetry (cylinder, cone), and spherical if the solid body has a centre of symmetry (sphere, hemisphere).

5.2 Cylindrical coordinates

The cylindrical coordinates for a point are the cylindrical *radius* ρ (distance from the z-axis), *azimuthal angle* or azimuth ϕ (angle between the projection of $\boldsymbol{r}$ onto the xy-plane and the positive direction of the x-axis), and the Cartesian z coordinate.



Their ranges are $0 \leq \rho < \infty$, $0 \leq \phi \leq 2\pi$, and $-\infty < z < +\infty$, and they are related to the Cartesian coordinates by[24]

$$x = \rho \cos\phi, \quad y = \rho \sin\phi, \quad z = z. \tag{5.12}$$

with the inverse formulae

$$\rho = \sqrt{x^2 + y^2}, \quad \tan\phi = \frac{y}{x}, \quad z = z. \tag{5.13}$$

From

$$\boldsymbol{r} = \rho \cos\phi\, \boldsymbol{i} + \rho \sin\phi\, \boldsymbol{j} + z\boldsymbol{k}, \tag{5.14}$$

we find the unit vectors in the directions in which a point moves when either ρ, or ϕ, or z increases:

$$\boldsymbol{e}_\rho = \frac{\partial \boldsymbol{r}/\partial \rho}{|\partial \boldsymbol{r}/\partial \rho|} = \cos\phi\, \boldsymbol{i} + \sin\phi\, \boldsymbol{j}, \tag{5.15a}$$

$$\boldsymbol{e}_\phi = \frac{\partial \boldsymbol{r}/\partial \phi}{|\partial \boldsymbol{r}/\partial \phi|} = -\sin\phi\, \boldsymbol{i} + \cos\phi\, \boldsymbol{j} \tag{5.15b}$$

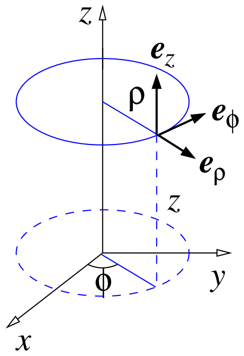
$$\boldsymbol{e}_z = \frac{\partial \boldsymbol{r}/\partial z}{|\partial \boldsymbol{r}/\partial z|} = \boldsymbol{k}, \tag{5.15c}$$

where $|\partial \boldsymbol{r}/\partial \rho| = 1$, $|\partial \boldsymbol{r}/\partial \phi| = \rho$ and $|\partial \boldsymbol{r}/\partial z| = 1$. The small displacements associated with increases in ρ, ϕ and z by $d\rho$, $d\phi$ and dz, respectively, are

$$(d\boldsymbol{r})_\rho = \frac{\partial \boldsymbol{r}}{\partial \rho} d\rho = \boldsymbol{e}_\rho d\rho, \tag{5.16a}$$

$$(d\boldsymbol{r})_\phi = \frac{\partial \boldsymbol{r}}{\partial \phi} d\phi = \boldsymbol{e}_\phi \rho d\phi, \tag{5.16b}$$

$$(d\boldsymbol{r})_z = \frac{\partial \boldsymbol{r}}{\partial z} dz = \boldsymbol{k} dz. \tag{5.16c}$$



It is easy to check, using dot product, that the unit vector $\boldsymbol{e}_\rho$ is orthogonal to $\boldsymbol{e}_\phi$, and they are both orthogonal to $\boldsymbol{e}_z = \boldsymbol{k}$. The same is true for the displacements (5.16a)–(5.16c). Hence, the volume element is given by the product of their lengths,

$$dV = \rho d\rho d\phi dz, \tag{5.17}$$

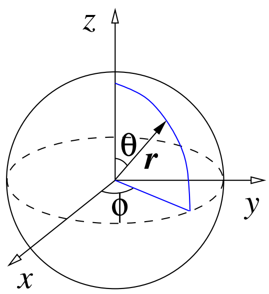
the squared length element is

$$ds^2 = d\rho^2 + \rho^2 d\phi^2 + dz^2, \tag{5.18}$$

and the element of the cylindrical surface (ρ = const) is obtained by multiplying the lengths of $(d\boldsymbol{r})_\phi$ and $(d\boldsymbol{r})_z$,

$$dS = \rho d\phi dz. \tag{5.19}$$

5.3 Spherical coordinates



The spherical coordinates for a point are the *radius* r (distance from the origin), *polar angle* θ (angle between $\boldsymbol{r}$ and positive direction of the z-axis), and *azimuthal angle* or azimuth ϕ (angle between the projection of $\boldsymbol{r}$ onto the xy-plane and the positive direction of the x-axis).

Their ranges are $0 \le r < \infty$, $0 \le \theta \le \pi$, and $0 \le \phi \le 2\pi$, and they are related to the Cartesian coordinates by

$$x = r\sin\theta\cos\phi, \quad y = r\sin\theta\sin\phi, \quad z = r\cos\theta. \tag{5.20}$$

[24] The cylindrical coordinates ρ and ϕ are similar to the plane polar coordinates r and θ, Sec. 4.4.

with the inverse formulae[25]

$$r = \sqrt{x^2 + y^2 + z^2}, \quad \theta = \arccos \frac{z}{\sqrt{x^2 + y^2 + z^2}}, \quad \tan\phi = \frac{y}{x}. \tag{5.21}$$

From

$$\boldsymbol{r} = r\sin\theta\cos\phi\,\boldsymbol{i} + r\sin\theta\sin\phi\,\boldsymbol{j} + r\cos\theta\,\boldsymbol{k},$$

we find the unit vectors in the directions in which a point moves when either r, or θ, or ϕ increases:

$$\boldsymbol{e}_r = \frac{\partial\boldsymbol{r}/\partial r}{|\partial\boldsymbol{r}/\partial r|} = \sin\theta\cos\phi\,\boldsymbol{i} + \sin\theta\sin\phi\,\boldsymbol{j} + \cos\theta\,\boldsymbol{k}, \tag{5.22a}$$

$$\boldsymbol{e}_\theta = \frac{\partial\boldsymbol{r}/\partial\theta}{|\partial\boldsymbol{r}/\partial\theta|} = \cos\theta\cos\phi\,\boldsymbol{i} + \cos\theta\sin\phi\,\boldsymbol{j} - \sin\theta\,\boldsymbol{k}, \tag{5.22b}$$

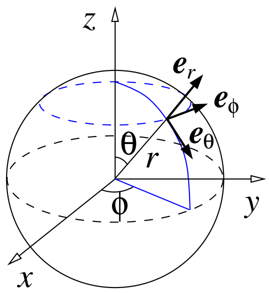
$$\boldsymbol{e}_\phi = \frac{\partial\boldsymbol{r}/\partial\phi}{|\partial\boldsymbol{r}/\partial\phi|} = -\sin\phi\,\boldsymbol{i} + \cos\phi\,\boldsymbol{j}, \tag{5.22c}$$

where $|\partial\boldsymbol{r}/\partial r| = 1$, $|\partial\boldsymbol{r}/\partial\theta| = r$ and $|\partial\boldsymbol{r}/\partial\phi| = r\sin\theta$. The small displacements associated with increases in r, θ and ϕ by dr, $d\theta$ and $d\phi$, respectively, are

$$(d\boldsymbol{r})_r = \frac{\partial\boldsymbol{r}}{\partial r}dr = \boldsymbol{e}_r dr, \tag{5.23a}$$

$$(d\boldsymbol{r})_\theta = \frac{\partial\boldsymbol{r}}{\partial\theta}d\theta = \boldsymbol{e}_\theta r d\theta, \tag{5.23b}$$

$$(d\boldsymbol{r})_\phi = \frac{\partial\boldsymbol{r}}{\partial\phi}d\phi = \boldsymbol{e}_\phi r\sin\theta d\phi. \tag{5.23c}$$



It is easy to check, using the dot product, that the unit vectors $\boldsymbol{e}_r$, $\boldsymbol{e}_\theta$ and $\boldsymbol{e}_\phi$, Eqs. (5.22a)–(5.22c), are mutually orthogonal. The same is true for the displacements (5.23a)–(5.23c). Hence, the associated volume element is found simply as the product of their lengths,

$$dV = r^2\sin\theta\,dr d\theta d\phi. \tag{5.24}$$

We also obtain the squared length element as

$$ds^2 = dr^2 + r^2 d\theta^2 + r^2\sin^2\theta d\phi^2, \tag{5.25}$$

and the element of the spherical surface $r = \text{const}$, by multiplying the lengths of $(d\boldsymbol{r})_\theta$ and $(d\boldsymbol{r})_\phi$,

$$dS = r^2\sin\theta\,d\theta d\phi. \tag{5.26}$$

[25] When finding ϕ from Eq. (5.21), use $\phi = \arctan(y/x)$ for $x > 0$, and $\phi = \arctan(y/x) + \pi$ (or $-\pi$) for $x < 0$.

5.4 Orthogonal curvilinear coordinates

In general, one can change from the Cartesian coordinates x, y, z, to some new coordinates q_1, q_2, q_3,

$$x = x(q_1, q_2, q_3), \quad y = y(q_1, q_2, q_3), \quad z = z(q_1, q_2, q_3), \tag{5.27}$$

or

$$\boldsymbol{r} = x(q_1, q_2, q_3)\boldsymbol{i} + y(q_1, q_2, q_3)\boldsymbol{j} + z(q_1, q_2, q_3)\boldsymbol{k}. \tag{5.28}$$

If we hold two of the new coordinates fixed and let the third one vary, the point $\boldsymbol{r}$ will describe a curve in space. In general, such curves will not be straight lines, hence, the coordinates are called *curvilinear*.

The displacements associated with small changes dq_i of each of the new coordinates ($i = 1$, 2, 3), are

$$(d\boldsymbol{r})_{q_1} = \frac{\partial \boldsymbol{r}}{\partial q_1} dq_1 = \left(\frac{\partial x}{\partial q_1} \boldsymbol{i} + \frac{\partial y}{\partial q_1} \boldsymbol{j} + \frac{\partial z}{\partial q_1} \boldsymbol{k} \right) dq_1, \tag{5.29}$$

$$(d\boldsymbol{r})_{q_2} = \frac{\partial \boldsymbol{r}}{\partial q_2} dq_2 = \left(\frac{\partial x}{\partial q_2} \boldsymbol{i} + \frac{\partial y}{\partial q_2} \boldsymbol{j} + \frac{\partial z}{\partial q_2} \boldsymbol{k} \right) dq_2, \tag{5.30}$$

$$(d\boldsymbol{r})_{q_3} = \frac{\partial \boldsymbol{r}}{\partial q_3} dq_3 = \left(\frac{\partial x}{\partial q_3} \boldsymbol{i} + \frac{\partial y}{\partial q_3} \boldsymbol{j} + \frac{\partial z}{\partial q_3} \boldsymbol{k} \right) dq_3. \tag{5.31}$$

If the vectors $(d\boldsymbol{r})_{q_1}$, $(d\boldsymbol{r})_{q_2}$ and $(d\boldsymbol{r})_{q_3}$ are mutually perpendicular, the coordinate system in known as *orthogonal*. In this case it is convenient to introduce the *Lamé* coefficients $h_i = |\partial \boldsymbol{r}/\partial q_i|$, so that we have

$$(d\boldsymbol{r})_{q_i} = \frac{\partial \boldsymbol{r}}{\partial q_i} dq_i = \underbrace{\frac{\partial \boldsymbol{r}/\partial q_i}{|\partial \boldsymbol{r}/\partial q_i|}}_{\text{unit vector } \boldsymbol{e}_{q_i}} |\partial \boldsymbol{r}/\partial q_i| dq_i = h_i \boldsymbol{e}_{q_i} dq_i. \tag{5.32}$$

This allows us to find the squared length element as

$$ds^2 = h_1^2 dq_1^2 + h_2^2 dq_2^2 + h_3^2 dq_3^2, \tag{5.33}$$

and the volume element as

$$dV = h_1 h_2 h_3 dq_1 dq_2 dq_3. \tag{5.34}$$

Examples. As seen in Sec. 5.3, in spherical coordinates r, θ, ϕ,

$$h_r = 1, \quad h_\theta = r, \quad h_\phi = r \sin\theta, \tag{5.35}$$

and in cylindrical coordinates ρ, ϕ, z (Sec. 5.2),

$$h_\rho = 1, \quad h_\phi = \rho, \quad h_z = 1. \tag{5.36}$$

If the curvilinear coordinates *are not* orthogonal, the vectors $(d\boldsymbol{r})_{q_1}$, $(d\boldsymbol{r})_{q_2}$ and $(d\boldsymbol{r})_{q_3}$ define a parallelepiped. The volume element dV is then obtained from the triple scalar product[26] $(d\boldsymbol{r})_{q_1} \cdot [(d\boldsymbol{r})_{q_2} \times (d\boldsymbol{r})_{q_2}]$, which can be written as a determinant,

$$dV = \begin{vmatrix} \dfrac{\partial x}{\partial q_1} & \dfrac{\partial y}{\partial q_1} & \dfrac{\partial z}{\partial q_1} \\ \dfrac{\partial x}{\partial q_2} & \dfrac{\partial y}{\partial q_2} & \dfrac{\partial z}{\partial q_2} \\ \dfrac{\partial x}{\partial q_3} & \dfrac{\partial y}{\partial q_3} & \dfrac{\partial z}{\partial q_3} \end{vmatrix} dq_1 dq_2 dq_3. \tag{5.37}$$

The above determinant is called the *Jacobian* and is denoted

$$J = \frac{D(x, y, z)}{D(q_1, q_2, q_3)}. \tag{5.38}$$

The volume element then is

$$dV = |J| dq_1 dq_2 dq_3, \tag{5.39}$$

where we use modulus to keep dV positive, irrespectively of the sign of J.

<u>Example.</u> The Jacobians for the spherical and cylindrical coordinates are

$$\frac{D(x, y, z)}{D(r, \theta, \phi)} = r^2 \sin\theta, \quad \text{and} \quad \frac{D(x, y, z)}{D(\rho, \phi, z)} = \rho.$$

Verify this using Eqs. (5.20) and (5.12).

[26] The triple scalar product of vectors $\boldsymbol{a}$, $\boldsymbol{b}$ and $\boldsymbol{c}$ gives the volume of a parallelepiped with edges given by these vectors, and can be evaluated from the determinant:

$$\boldsymbol{a} \cdot (\boldsymbol{b} \times \boldsymbol{c}) = \begin{vmatrix} a_x & a_y & a_z \\ b_x & b_y & b_z \\ c_x & c_y & c_z \end{vmatrix}.$$

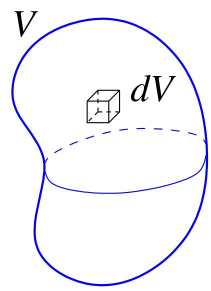
6 Volume and surface integrals

6.1 Volume integrals

We know that we can use triple integrals to calculate the volumes of 3D shapes. From Chapter 5 the volume of a closed, bounded region R in space is

$$V = \iiint_R dV$$

Consider a solid object made from a material of constant density γ[27]. If we know the volume V of the body, then its mass is $M = \gamma V$. How can we find the mass of the body if the density is not constant, i.e., $\gamma = f(x, y, z)$?



Let us consider a small volume element dV within which the density is constant. Its mass is $dm = \gamma dV$, and the total mass of the object can be found by adding all such contributions over the whole volume of the body,

$$M = \int\limits_V dm = \int\limits_V \gamma dV. \tag{6.1}$$

If we use Cartesian coordinates, $dV = dxdydz$, this integral becomes a triple integral of the form

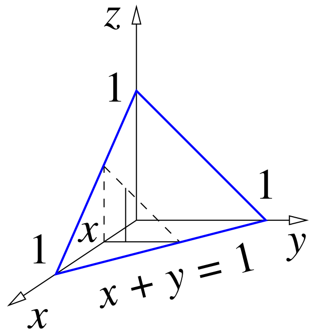
$$\iiint\limits_V f(x, y, z) dxdydz. \tag{6.2}$$

This is a *volume integral*. To evaluate it, we need to know the function f and the shape of the body, which determines the integration limits.

<u>Example.</u> Evaluate $\displaystyle\iiint\limits_V \frac{dxdydz}{\sqrt{x+y+z+1}}$, where V is bounded by the coordinate planes $(x = 0,\ y = 0$ and $z = 0)$ and the plane $x + y + z = 1$.

The volume in question is a tetrahedron (triangular pyramid) with three vertices in the x-y plane: $(0, 0, 0)$, $(1, 0, 0)$ and $(0, 1, 0)$, and one vertex above it: $(0, 0, 1)$. For a given x

[27] Density is mass per unit volume.



between 0 and 1, the range of y is from 0 to the line $x+y=1$ in the xy-plane, and for a given x and y, z varies between 0 and the plane $x+y+z=1$. Hence, the integration region is defined by

$$0 \leq x \leq 1, \quad 0 \leq y \leq 1-x, \quad 0 \leq z \leq 1-x-y,$$

and the integral can be written as

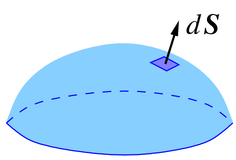
$$\int_0^1 \int_0^{1-x} \int_0^{1-x-y} \frac{dzdydx}{\sqrt{x+y+z+1}} = \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} \frac{1}{\sqrt{x+y+z+1}} dz$$
$$= \int_0^1 dx \int_0^{1-x} dy \left[2\sqrt{x+y+z+1} \right]_0^{1-x-y} = 2\int_0^1 dx \int_0^{1-x} dy \left[\sqrt{2} - \sqrt{x+y+1} \right]$$
$$= 2\int_0^1 dx \left[\sqrt{2}y - \frac{2}{3}(x+y+1)^{3/2} \right]_0^{1-x} = 2\int_0^1 dx \left[-\frac{\sqrt{2}}{3} - \sqrt{2}x + \frac{2}{3}(x+1)^{3/2} \right]$$
$$= 2\left[-\frac{\sqrt{2}}{3}x - \frac{\sqrt{2}}{2}x^2 + \frac{2}{3}\frac{2}{5}(x+1)^{5/2} \right]_0^1 = \frac{7\sqrt{2}-8}{15}.$$

6.2 Surface integrals

In this section we will learn how to integrate both scalar-valued functions and vector fields along surfaces in $\mathbb{R}^3$. Consider a surface with a surface area element dS. It is convenient to regard the surface element as a vector

$$d\boldsymbol{S} = \boldsymbol{n} dS, \tag{6.3}$$

where $\boldsymbol{n}$ is a unit normal vector. According to this definition, $d\boldsymbol{S}$ is perpendicular to the surface and $|d\boldsymbol{S}| = dS$.



A surface integral of the first kind is

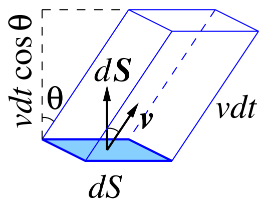
$$\int_S f(x,y,z)dS. \tag{6.4}$$

It can be used to find the surface area itself (by setting $f = 1$), or the mass of the surface, if $f(x,y,z)$ is the area mass density (i.e., mass per unit area).

A surface integral of the second kind is

$$\int_S \boldsymbol{A} \cdot d\boldsymbol{S}, \tag{6.5}$$

where $\boldsymbol{A}$ is a vector function.



The integral (6.5) is known as the *flux* of vector $\boldsymbol{A}$ across the surface S. To understand its meaning, consider the flow of a fluid across a small surface element dS. Let the fluid velocity $\boldsymbol{v}$ make angle θ with $d\boldsymbol{S}$. The distance travelled by the fluid in time dt is vdt. Hence, the volume dV of the fluid that flows across dS in time dt is given by the volume of a parallelepiped,

$$dV = dSvdt\cos\theta = \boldsymbol{v} \cdot d\boldsymbol{S}dt.$$

The volume that flows across dS in unit time is then $dV/dt = \boldsymbol{v} \cdot d\boldsymbol{S}$, and the volume that flows across the whole surface S in unit time (i.e., the flux) is

$$\int_S \boldsymbol{v} \cdot d\boldsymbol{S}.$$

<u>Example 1.</u> Evaluate $\int_S \boldsymbol{A} \cdot d\boldsymbol{S}$ for $\boldsymbol{A} = x\boldsymbol{i} + y\boldsymbol{j} + z\boldsymbol{k}$, if S is a rectangle,

$$0 \le x \le 2, \quad 0 \le y \le 3, \quad z = 1,$$

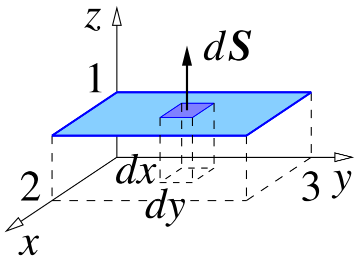
and $d\boldsymbol{S}$ is in the positive direction of the z-axis.

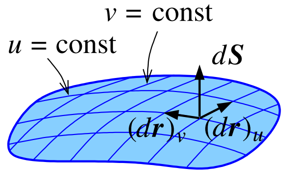
The rectangle is parallel to the xy-plane, so $d\boldsymbol{S} = \boldsymbol{k}dxdy$, and

$$\boldsymbol{A} \cdot d\boldsymbol{S} = (x\boldsymbol{i} + y\boldsymbol{j} + z\boldsymbol{k}) \cdot \boldsymbol{k}dxdy = zdxdy = dxdy,$$

since $z = 1$ on S. Hence,

$$\int_S \boldsymbol{A} \cdot d\boldsymbol{S} = \int_0^3 \int_0^2 dxdy = \int_0^3 dy \int_0^2 dx = 6.$$





In general, for a surface parameterised by $\boldsymbol{r} = \boldsymbol{r}(u, v)$, the displacement vectors corresponding to the change in u by du, and v by dv, are

$$(d\boldsymbol{r})_u = \frac{\partial \boldsymbol{r}}{\partial u} du, \quad \text{and} \quad (d\boldsymbol{r})_v = \frac{\partial \boldsymbol{r}}{\partial v} dv, \tag{6.6}$$

along the lines $v = \text{const}$ and $u = \text{const}$, respectively, and the surface element is[28]

$$d\boldsymbol{S} = (d\boldsymbol{r})_u \times (d\boldsymbol{r})_v = \frac{\partial \boldsymbol{r}}{\partial u} \times \frac{\partial \boldsymbol{r}}{\partial v} dudv, \tag{6.7}$$

with magnitude[29]

$$dS = \left| \frac{\partial \boldsymbol{r}}{\partial u} \times \frac{\partial \boldsymbol{r}}{\partial v} \right| dudv. \tag{6.8}$$

Example 2. Find the area of a surface given by the equation $z = z(x, y)$.

This surface is described by

$$\boldsymbol{r} = x\boldsymbol{i} + y\boldsymbol{j} + z(x, y)\boldsymbol{k},$$

where x and y can be viewed as the two parameters. Hence,

$$\frac{\partial \boldsymbol{r}}{\partial x} = \boldsymbol{i} + \frac{\partial z}{\partial x}\boldsymbol{k}, \quad \frac{\partial \boldsymbol{r}}{\partial y} = \boldsymbol{j} + \frac{\partial z}{\partial y}\boldsymbol{k},$$

$$d\boldsymbol{S} = \frac{\partial \boldsymbol{r}}{\partial x} \times \frac{\partial \boldsymbol{r}}{\partial y} dxdy = \begin{vmatrix} \boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k} \\ 1 & 0 & \dfrac{\partial z}{\partial x} \\ 0 & 1 & \dfrac{\partial z}{\partial y} \end{vmatrix} dxdy = \left(-\frac{\partial z}{\partial x}\boldsymbol{i} - \frac{\partial z}{\partial y}\boldsymbol{j} + \boldsymbol{k} \right) dxdy,$$

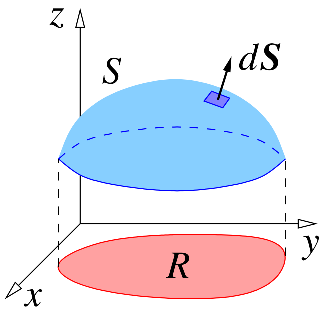
[28] An alternative choice is $d\boldsymbol{S} = -(\partial \boldsymbol{r}/\partial u) \times (\partial \boldsymbol{r}/\partial v) dudv = (\partial \boldsymbol{r}/\partial v) \times (\partial \boldsymbol{r}/\partial u) dudv$.

[29] Recall that $\boldsymbol{a} \times \boldsymbol{b}$ is perpendicular to both vectors $\boldsymbol{a}$ and $\boldsymbol{b}$, and $|\boldsymbol{a} \times \boldsymbol{b}|$ is the area of a parallelogram with sides $\boldsymbol{a}$ and $\boldsymbol{b}$.

$$dS = |d\boldsymbol{S}| = \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2}\, dxdy,$$

and the area of the surface S over some region R in the xy-plane is

$$A = \int_S dS = \iint_R \sqrt{1 + (\partial z/\partial x)^2 + (\partial z/\partial y)^2}\, dxdy. \tag{6.9}$$



Let us use Eq. (6.9) to find the area of the hemisphere of $z = \sqrt{a^2 - x^2 - y^2}$ (see Sec. 2). We have:

$$A = \iint_R \left[1 + \frac{x^2}{a^2 - x^2 - y^2} + \frac{y^2}{a^2 - x^2 - y^2}\right]^{1/2} dxdy = \iint_R \frac{adxdy}{\sqrt{a^2 - x^2 - y^2}},$$

where R is the disc $x^2 + y^2 \le a^2$. Integration is easier if we use polar coordinates, replacing $dxdy$ by $rdrd\theta$ [see Eq. (4.16)],

$$A = \int_0^{2\pi}\int_0^a \frac{ardrd\theta}{\sqrt{a^2 - r^2}} = -\frac{a}{2}\int_0^{2\pi} d\theta \int_0^a \frac{d(a^2 - r^2)}{\sqrt{a^2 - r^2}} = -2\pi a\left[\sqrt{a^2 - r^2}\right]_0^a = 2\pi a^2,$$

which is a half of $4\pi a^2$, the surface area of a sphere of radius a.

Example 3. Evaluate $\oint_S \boldsymbol{A}\cdot d\boldsymbol{S}$, where $\boldsymbol{A} = x\boldsymbol{i} + y\boldsymbol{j} + z\boldsymbol{k}$, S is an ellipsoid, $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} + \dfrac{z^2}{c^2} = 1$, with $d\boldsymbol{S}$ pointing outwards (i.e., away from the origin).

The ellipsoid can be parameterised by

$$x = a\sin u\cos v, \quad y = b\sin u\sin v, \quad z = c\cos u,$$

where $0 \le u \le \pi$ and $0 \le v \le 2\pi$ [cf. (5.4)] (verify!). Hence,

$$\boldsymbol{r} = a\sin u\cos v\boldsymbol{i} + b\sin u\sin v\boldsymbol{j} + c\cos u\boldsymbol{k},$$

and

$$\frac{\partial \boldsymbol{r}}{\partial u} = a\cos u\cos v\boldsymbol{i} + b\cos u\sin v\boldsymbol{j} - c\sin u\boldsymbol{k}$$
$$\frac{\partial \boldsymbol{r}}{\partial v} = -a\sin u\sin v\boldsymbol{i} + b\sin u\cos v\boldsymbol{j},$$

so that the surface element (6.7) is

$$d\boldsymbol{S} = \begin{vmatrix} \boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k} \\ a\cos u \cos v & b\cos u \sin v\boldsymbol{j} & -c\sin u \\ -a\sin u\sin v & b\sin u\cos v & 0 \end{vmatrix} dudv$$
$$= (bc\sin^2 u\cos v\boldsymbol{i} + ac\sin^2 u\sin v\boldsymbol{j} + ab\sin u\cos u\boldsymbol{k})dudv,$$

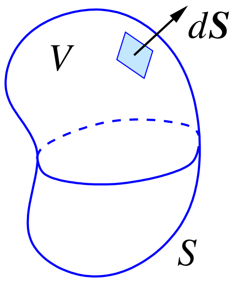
and

$$\begin{aligned} \boldsymbol{A}\cdot d\boldsymbol{S} &= (xbc\sin^2 u\cos v + yac\sin^2 u\sin v + zab\sin u\cos u)dudv \\ &= (abc\sin^3 u\cos^2 v + bac\sin^3 u\sin^2 v + cab\sin u\cos^2 u)dudv \\ &= abc\sin u\, dudv, \end{aligned}$$

The integral then is

$$\oint_S \boldsymbol{A}\cdot d\boldsymbol{S} = \int_0^{2\pi}\int_0^{\pi} abc\sin u\, dudv = abc\int_0^{2\pi} dv\int_0^{\pi}\sin u\, du = 4\pi abc.$$

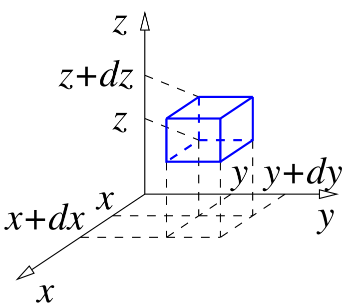
6.3 Gauss's theorem



Gauss's theorem (also known as the divergence theorem) states:

$$\oint_S \boldsymbol{A}\cdot d\boldsymbol{S} = \int_V \nabla\cdot\boldsymbol{A}\, dV, \tag{6.10}$$

where S is a closed surface, V is the volume enclosed by S, and $d\boldsymbol{S}$ is outwards from the volume V.



Proof. To prove Gauss's theorem, consider a small rectangular prism with faces parallel to the coordinate planes, at x, $x+dx$, y, $y+dy$, z and $z+dz$. The corresponding surface integral

$$\oint_{\square} \boldsymbol{A} \cdot d\boldsymbol{S}$$

consists of six contributions, one from each of the faces. For the face at x, the surface element is $d\boldsymbol{S} = -\boldsymbol{i}dydz$ and its contribution to the above integral is

$$\boldsymbol{A} \cdot d\boldsymbol{S} = -A_x(x)dydz,$$

omitting other variables in A_x. For the face at $x+dx$, $d\boldsymbol{S} = \boldsymbol{i}dydz$, and its contribution is

$$A_x(x+dx)dydz \simeq \left[A_x(x) + \frac{\partial A_x}{\partial x}dx\right] dydz,$$

where we approximated $A_x(x+dx)$ using Taylor's expansion in dx to first order. The contribution of these two faces then is

$$\frac{\partial A_x}{\partial x}dxdydz.$$

Similarly, the contribution of the two faces at y and dy is

$$\frac{\partial A_y}{\partial y}dxdydz,$$

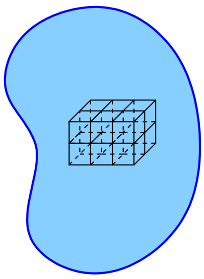
and that of the two faces at z and $z+dz$ is

$$\frac{\partial A_z}{\partial z}dxdydz.$$

Hence, the total flux across the surface S of the small rectangular prism is

$$\oint_{\square} \boldsymbol{A} \cdot d\boldsymbol{S} = \left(\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}\right) dxdydz = \nabla \cdot \boldsymbol{A}\, dxdydz = \int_{\square} \nabla \cdot \boldsymbol{A}\, dV,$$

where $dxdydz$ is the volume of the prism, and where we also used the fact that $\nabla \cdot \boldsymbol{A}$ can be regarded as constant, since the prism is small.



A large volume V can be "diced" into small rectangular prisms V_i with surfaces S_i. From the above, Gauss's theorem is valid for each of them. Adding their contributions over all V, we have

$$\sum_i \oint_{S_i} \boldsymbol{A} \cdot d\boldsymbol{S} = \sum_i \int_{V_i} \nabla \cdot \boldsymbol{A} dV. \tag{6.11}$$

On the left-hand side, the fluxes across adjacent faces of neighbouring prisms cancel (because $d\boldsymbol{S}$ has opposite directions on them), leaving us with the integral over the surface S that bounds V, as on the left-hand side of Eq. (6.10). The right-hand side of Eq. (6.11) is simply the integral over the whole volume V, giving the right-hand side of Gauss's theorem (6.10).

<u>Example.</u> Show that the surface integral

$$\frac{1}{3} \oint_S \boldsymbol{r} \cdot d\boldsymbol{S} \tag{6.12}$$

gives the volume enclosed by S.

Indeed, by Gauss's theorem,

$$\frac{1}{3} \oint_S \boldsymbol{r} \cdot d\boldsymbol{S} = \frac{1}{3} \int_V \nabla \cdot \boldsymbol{r} dV = \int_V dV = V,$$

since $\nabla \cdot \boldsymbol{r} = 3$.

Let us use the integral (6.12) to find the volume of the right circular cone of height h and base radius a. Placing the apex of the cone at the origin, with its axis of symmetry along z (see example in Sec. 6.1), we see that $d\boldsymbol{S}$ is perpendicular to $\boldsymbol{r}$ everywhere on the conical surface, so that $\boldsymbol{r} \cdot d\boldsymbol{S} = 0$ there. Hence, only the base contributes to the flux in (6.12). On the base, $d\boldsymbol{S} = \boldsymbol{k} dS$ and $\boldsymbol{r} \cdot d\boldsymbol{S} = \boldsymbol{r} \cdot \boldsymbol{k} dS = z dS = h dS$, so that

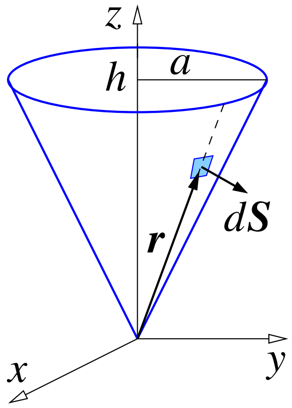
$$V = \frac{1}{3} \oint_S \boldsymbol{r} \cdot d\boldsymbol{S} = \frac{1}{3} \int_{\text{base}} \boldsymbol{r} \cdot d\boldsymbol{S} = \frac{1}{3} h \int_{\text{base}} dS = \frac{1}{3} \pi a^2 h.$$

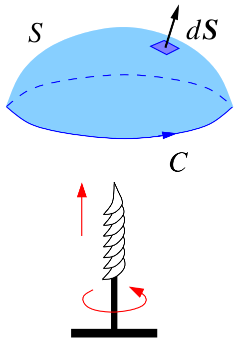
We can also use (6.12) to find the volume of the ellipsoid $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} + \dfrac{z^2}{c^2} = 1$. In Sec. 6.2, example 3, we evaluated the surface integral for $\boldsymbol{A} = \boldsymbol{r}$ for it, and found $\oint_S \boldsymbol{A} \cdot d\boldsymbol{S} = 4\pi abc$. Hence, the volume of the ellipsoid is $V = \dfrac{4\pi}{3} abc$. In particular, for $a = b = c \equiv R$, this gives the volume of the sphere $4\pi R^3/3$.

6.4 Stokes's theorem

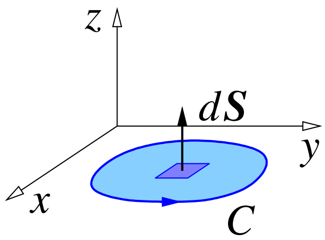
Stokes's theorem (also known as the circulation theorem) states:

$$\oint_C \boldsymbol{A} \cdot d\boldsymbol{r} = \int_S \nabla \times \boldsymbol{A} \cdot d\boldsymbol{S}, \tag{6.13}$$





where C is a closed curve which bounds surface S, and the direction in which C is traversed is linked with the direction of $d\boldsymbol{S}$ by the right-handed corkscrew rule[30].



Proof. Let us first consider a plane curve C spanned by a flat surface S, and choose the coordinates so that the curve lies in the xy-plane, and is traversed in the anticlockwise direction. In this case, the left-hand side of (6.13) is

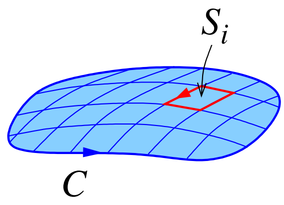
$$\oint_C \boldsymbol{A} \cdot d\boldsymbol{r} = \oint_C (A_x dx + A_y dy).$$

The surface element on the right-hand side of (6.13) is $d\boldsymbol{S} = \boldsymbol{k} dxdy$, so that

$$\int_S \nabla \times \boldsymbol{A} \cdot d\boldsymbol{S} = \iint_S (\nabla \times \boldsymbol{A})_z dxdy = \iint_S \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) dxdy,$$

and Eq. (6.13) follows from Green's theorem (4.18) for $P = A_x$ and $Q = A_y$.

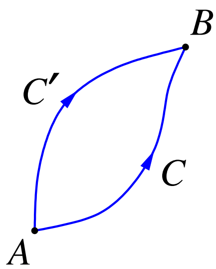
[30] If the corkscrew is rotated in the direction set on C, it moves in the direction of the surface normal $d\boldsymbol{S}$.



If C is not a plane curve and/or the surface S is not flat, the latter can be divided into small segments S_i, each with boundary C_i, which are approximately flat. Hence, Stoke's theorem is valid for each S_i, and we have

$$\sum_i \oint_{C_i} \boldsymbol{A} \cdot d\boldsymbol{r} = \sum_i \int_{S_i} \nabla \times \boldsymbol{A} \cdot d\boldsymbol{S}, \tag{6.14}$$

where the sum is over all segments that comprise S. On the left-hand side of (6.14), the line integrals along common boundaries of neighbouring segments cancel (because $d\boldsymbol{r}$ have opposite directions on them), leaving us with the integral over the curve C that bounds S, as on the left-hand side of Eq. (6.13). The right-hand side of Eq. (6.14) is simply the integral over the whole surface S, giving the right-hand side of Stokes's theorem. Stokes's theorem allows one to show that the line integral of an irrotational vector field between any two points is independend of the path taken. This fact was used in Sec. 4.2 when finding the potential for an irrotational field.



Let points A and B be connected by curves C and C'. We want to show that for $\nabla \times \boldsymbol{A} = 0$,

$$\int_{ACB} \boldsymbol{A} \cdot d\boldsymbol{r} = \int_{AC'B} \boldsymbol{A} \cdot d\boldsymbol{r}, \tag{6.15}$$

where the integral on the left-hand side is from A to B along C, and that on the right-hand side is from A to B along C'. Consider a closed curve $ACBC'A$ that follows C from A to B, and follows C' from B to A. By Stokes's theorem,

$$\oint_{ACBC'A} \boldsymbol{A} \cdot d\boldsymbol{r} = \int_S \nabla \times \boldsymbol{A} \cdot d\boldsymbol{S} = 0,$$

where S is any surface bounded by the closed curve $ACBC'A$. Therefore,

$$\oint_{ACBC'A} \boldsymbol{A} \cdot d\boldsymbol{r} = \int_{ACB} \boldsymbol{A} \cdot d\boldsymbol{r} + \int_{BC'A} \boldsymbol{A} \cdot d\boldsymbol{r} = \int_{ACB} \boldsymbol{A} \cdot d\boldsymbol{r} - \int_{AC'B} \boldsymbol{A} \cdot d\boldsymbol{r} = 0,$$

where we used Eq. (4.5) in the second step, which proves (6.15).